



Documaker

Docutoolbox Reference

version 11.3

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Utility Reference

This manual contains information about the various Docutoolbox utilities you can use.

Refer to the [Finding the Right Utility on page 2](#) to quickly locate the utility you want to use. This list includes a short description of each utility.

The remainder of this manual describes each utility, discusses the parameters you can set, and provides examples.

The utilities are discussed in alphabetical order.

NOTE The name *Docutoolbox* refers to the tools you use to create the form sets and resources required for your Documaker solution. This manual discusses the command-line utilities that are a part of Docutoolbox.

FINDING THE RIGHT UTILITY

Use this table to locate the utility you want more information about. All utilities are discussed on the following pages in alphabetical order.

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ADDCRLF Use the ADDCRLF utility to prepare AFP resource files created in a Windows environment for uploading to MVS systems. This utility adds carriage return/line feeds (CR/LF) to the files. The CR/LFs serve as record delimiters.

NOTE Some AFP resources, such as AFPBAT1 output from the GenPrint program, can have carriage return/line feeds (x'0D0A') embedded in them. On these resources, the ADDCRLF utility will fail, giving you a message stating the data has embedded carriage return/line feeds (CR/LFs). If this happens, do the file transfer with *no CR/LF* and then use the AFP2MVS utility on the MVS system. Refer to [AFP2MVS on page 67](#) for more information.

Program names

Windows ADDCRLF.EXE

Syntax ADDCRLF /I /F

Parameter Description

/I	The AFP file name with the extension TMP, which you can omit.
/F	Add this parameter to force the output file to be built, ignoring any embedded CR/LF errors.

For the input file, the utility looks for the default extension **TM**P. For the output file, the utility will assign the extension **IB**M.

PSF software on the MVS is record-oriented software. It needs each structured field record in an AFP resource such as font, formdef, overlay, page segment, and so on, in discrete records.

PSF2 on a Windows system has no concept of records. The entire datastream is a single record. If you use the system in a Windows environment to create AFP resources, and you want to transfer them to MVS, *you must* insert CR/LFs at the end of each structured field and then use the CR/LF keyword on the file upload command.

Example ADDCRLF /I=afpfile

This creates a file called AFPFILE.IBM. The input file must have the extension **TM**P. You do not have to enter the extension if you enter the command as shown above. The output file will have the same file name, but with the **IB**M extension.

AFP2MVS Use the AFP2MVS utility to *record-orient* an AFP file that has been transferred to z/OS (MVS, OS/390) from a Windows or UNIX-based system.

When an AFP file such as a print stream, font, or a page segment is uploaded from Windows or UNIX to a z/OS (OS/390, MVS) system, the resulting AFP file is not oriented into separate records — instead it looks like one continuous record. For the AFP file to be properly processed on z/OS, it must be oriented into separate records.

Because of the structure of AFP records, these records can be separated fairly easily. Each AFP record usually begins with a 0x5A byte. In the EBCDIC character set, this value (code point) is displayed as an exclamation point (!). Following the 0x5A byte is the length of the record. The AFP2MVS utility reads through the AFP file, separates the file into records and writes the resulting record-oriented file out as an output file. If you were then to browse the file, the 0x5A records (first byte displayed as an exclamation point) would display one per row.

Program names

z/OS (MVS) AFP2MVS

Syntax AFP2MVS [/I=] [/L=]

Parameter Description

/I	If the input file is a Partitioned Data Set (PDS), use this parameter to specify the member within the PDS to process. To process all members, enter /I=*. If the input file is a sequential file, omit this parameter.
/L	Include this parameter to tell the utility that the dataset is a Mixed Mode dataset. Unlike regular AFP files, Mixed Mode AFP files typically contain records of line data — not AFP records

Keep in mind...

- Sample JCL for this utility is located in member AFP2MV SX in the JCLLIB PDS that is created when Documaker Server is installed on z/OS.
- In Documaker Server version 11.0 and earlier, the DD names RSCOS2 and RSCMVS were used to designate the input and output files. The preferred DD names are now *INPUT* and *OUTPUT*.
- When you transfer AFP files from Windows or UNIX to z/OS, be sure to transfer the files in binary mode, not text mode.
- The dataset you create on z/OS to contain the AFP files you upload should have DCB characteristics similar to the dataset the AFP2MVS utility will write to. For example, for a sequential file you could specify DCB characteristics as shown here:

```
Data Set Name . . . . : FSI.V112.RPEX1.GENPRINT.PRTBAT1.FROMPC
```

General Data	Current Allocation
Management class . . . : **None**	Allocated cylinders : 1
Storage class : STANDARD	Allocated extents . : 1
Volume serial : DCI009	
Device type : 3390	
Data class : **None**	Current Utilization
Organization : PS	Used cylinders . . : 0
Record format : VBM	Used extents . . . : 0

```

Record length . . . : 8209
Block size . . . . : 23500
1st extent cylinders: 1
Secondary cylinders : 1
Data set name type :                               SMS Compressible . : NO

```

Example Here are some examples:

Example 1 This example demonstrates running the AFP2MVS utility on a print stream:

In this example, the AFP2MVS utility reads the print stream associated with the INPUT DD statement (FSI.V112.RPEX1.GENPRINT.PRTBAT1.FROMPC), separates the print stream into records, then writes the new print stream to the file associated with the OUTPUT DD statement (FSI.V112.RPEX1.GENPRINT.PRTBAT1).

Since the file associated with the INPUT DD statement is a Sequential Data Set and not a Partitioned Data Set (PDS), the //I parameter is not necessary.

```

//AFP2MVSD EXEC PGM=IEFBR14
//OUTPUT DD DSN=FSI.V112.RPEX1.GENPRINT.PRTBAT1,
//          DISP=(MOD,DELETE),SPACE=(TRK,0),UNIT=SYSDA
// *
//AFP2MVS EXEC PGM=AFP2MVS
//STEPLIB DD DSN=FSI.V112..LINKLIB,DISP=SHR
//          DD DSN=SYS1.SCEERUN,DISP=SHR
//SYSPRINT DD SYSOUT=*
//INPUT DD DSN=FSI.V112.RPEX1.GENPRINT.PRTBAT1.FROMPC,DISP=SHR
//OUTPUT DD DSN=FSI.V112.RPEX1.GENPRINT.PRTBAT1,
//          UNIT=SYSDA,SPACE=(CYL,(1,1)),DISP=(,CATLG),
//          DCB=(RECFM=VBM,LRECL=8209,BLKSIZE=23500)

```

Example 2 This example demonstrates running the AFP2MVS utility on a PDS that contains several fonts:

In this example, the AFP2MVS utility reads each member of the PDS that is associated with the INPUT DD statement (FSI.V112.RPEX1.FONTLIB.FROMPC). Each member is then separated into records and the resulting record-oriented member is written to the PDS associated with the OUTPUT DD statement (FSI.V112.RPEX1.FONTLIB).

Since the file associated with the INPUT DD statement is a PDS, the //I parameter is included to indicate which members to process. In this case, you specify //I=* to indicate that all members are to be processed.

NOTE In the EXEC statement's PARM parameter, the first forward slash (/) is used to separate any Language Environment (LE) runtime options from the program's command line options. If you need to specify any LE runtime options, place those options before the first forward slash. Any options placed after the first forward slash are considered options for the program being executed.

Documaker-related programs typically use the forward slash (/) or dash (-) as the initial character of a command line parameter, so the number of forward slashes on the command line will total one more than the number of parameters you specified.

```

//AFP2MVSD EXEC PGM=IEFBR14
//OUTPUT DD DSN=FSI.V112.RPEX1.FONTLIB,

```

```
//          DISP=(MOD,DELETE),SPACE=(TRK,0),UNIT=SYSDA
// *
//AFP2MVS EXEC PGM=AFP2MVS,PARM='/I=*'
//STEPLIB DD DSN=FSI.V112.LINKLIB,DISP=SHR
//          DD DSN=SYS1.SCEERUN,DISP=SHR
//SYSPRINT DD SYSOUT=*
//INPUT    DD DSN=FSI.V112.RPEX1.FONTLIB.FROMPC,DISP=SHR
//OUTPUT DD DSN=FSI.V112.RPEX1.FONTLIB,
//          UNIT=SYSDA,SPACE=(CYL,(1,1,1)),DISP=(,CATLG),
//          DSNTYPE=LIBRARY,
//          DCB=(RECFM=VBM,LRECL=8209,BLKSIZE=23500)
```

AFP2PCL Use the AFP2PCL utility to convert an AFP bitmap font into a corresponding PCL bitmap font. See following page for more information on AFP fonts and standard font naming conventions.

NOTE You can also perform this task using the Font Manager. See [Docucreate User Guide](#) for more information.

Program names

Windows AFP2PCLW.EXE

Syntax AFP2PCLW /I /S /D /O

Parameter Description

/I	The AFP file name, such as FILENAME.FNT (AFP coded font file), the extension is optional and will default to <i>.FNT</i> .
/S	The source dots per inch (DPI), either 240 or 300 (the AFP font DPI setting default is 300).
/D	The destination dots per inch, either 240 or 300 (the PCL font DPI setting default is 300).
/O	The output file, such as FILENAME.PCL (a PCL font file) If you omit the output name, the output PCL file name will be the same as the input file name, except for the PCL extension.

The utility creates a PCL character set file using codepage 1004. You must include the /I parameter.

Example AFP2PCL /I=x0facob9

This will convert an AFP coded font file *x0facob9.fnt* into a PCL font file named *x0facob9.pcl*

The AFP2PCL utility requires these files:

- AFP coded font file (system coded font files are named *X0DAxxxx.fnt*)
- AFP character set file (system character set files are named *C0FAxxxx.240/300*)
- AFP codepage file (the system's standard codepage is T1DOC037). The codepage file *cannot* have an extension.

In addition, AFP2PCL uses the FSISYS.INI and CODEPAGE.INI files. AFP2PCL looks for the FMRES control group in the FSISYS.INI file. In the FMRES control group, AFP2PCL looks for DefLib and CodePage options. Set the DefLib option to the directory where your system resources are stored.

Set the CodePage option to the name of the INI file which contains codepage information. The system looks for the CODEPAGE.INI file in the DefLib path specified in the FMRES control group. In this example,

```
< FMRes >
CodePage = CODEPAGE.INI
```

```
DefLib      = \newfonts\agfa\
```

the name of the INI file containing codepage information `CODEPAGE.INI` will be located in the `\newfonts\agfa` directory.

AFP terminology

An AFP font is composed of these component files: *coded font*, *codepage*, and *character set files*. The coded font file contains the names of character set and codepage files to use when printing.

NOTE You can use the AFPDUMP utility to find out the names of character set and codepage files contained in a coded font file. [See DUMP on page 10](#) for more information.

The *codepage* maps text to the characters in a character set file. Each character, like a capital 'A', has a particular name and a particular numeric value known as a *code point*. The character capital 'A' has an AFP name of LA020000. On a PC, a capital 'A' usually has a code point of 65. However, on a mainframe, a capital 'A' usually has a code point of 193.

For both the PC and the mainframe to print the letter 'A', a different codepage may be used. The codepage used to print text from the PC would associate a code point of 65 with the letter 'A' (LA020000). The codepage used to print text from the mainframe would associate a code point of 193 with the letter 'A' (LA020000). The codepage is merely a list of the character names and their associated code points.

The *character set file* contains the characters which can be printed. Like PCL fonts, each character set file can only represent a single font typeface, style, and point size. A character set file also specifies the list of AFP character names, like LA020000, it can print.

So, when you try to print the letter 'A' (code point 65 on a PC) using an AFP coded font file, the system examines the coded font file to determine the names of the codepage and character set files. It then looks up code point 65 in the codepage file to find the AFP character name associated with it. If the codepage is set up for PC printing, it finds the character named LA020000 associated with code point 65. The character set file is then used to print the bitmap information associated with the character name LA020000. An AFP character name, like LA020000, must be present in both the AFP codepage and character set files.

Font naming convention

The system's AFP coded font files are named X0DAxxxx.FNT, where **X0** indicates a coded font file, **D** symbolizes Documaker, and **A** denotes the AFP font. The first two **xx**s indicate the font type family. The next **x** indicates style. The last **x** indicates the font point size.

AFP character set files are named C0FAxxxx.240 or C0FAxxxx.300. **C0** indicates a character set file. The extension **240** and **300** indicates the dots per inch or DPI setting. The rest of the naming convention is the same as previously discussed.

AFP2VB Use this utility if you have a printable (native) AFP file and you want to view it to use it with Docuview LFS. This utility converts the file into variable block format.

Program names

Windows AFP2VB.EXE

Syntax

AFP2VB /I=inputfile /O=outputfile /D

Parameter Description

/I	The name of the native AFP file.
/O	The name of the output AFP file which will have variable block record format.
/D	(Optional) This parameter tells the utility to add Docusave comments.

AFPCF Use the AFPCF utility to create coded font files for AFP printers.

Program names

Windows AFPCFW32.EXE

Syntax AFPCFW32 /C /T /X

Parameter Description

/C	The AFP character set file name.
/T	The AFP codepage file name.
/X	The AFP codefont file to create (.FNT).

Example You would enter:

```
AFPCFW32 /c=C0FATIN0 /t=T1DOC037 /x=X0DATIN0
```

In this example, AFPCF creates an AFP coded font file called X0DATIN0.FNT.

To print using this coded font, all of these files must be resident on the printer. On MVS, it should be placed in a PDS which is available to the PSF procedure. On a Novell print server, it must be installed into the coded font group within PSF2.

To use this coded font for printing with Documaker Server, enter the name of the coded font file (without the FNT extension) into the Coded Font File field on the Printers tab of the Font Maintenance window in the Font Manager. For more information about the Font Manager, see the [Docucreate User Guide](#)

AFPCOPY Use the AFPCOPY utility to convert a print stream from *front-sheet* to *continuous-form*. The utility does this by taking an AFP print stream which contains multiple BEGINDOC and ENDDOC statements and generating a print stream with only one BEGINDOC and one ENDDOC statement.

Program names

Windows	AFPCOPY.EXE
MVS	AFPCOPY

Syntax AFPCOPY /I /O

Parameter Description

/I	The input file name (with page breaks).
/O	The output file name.

No parameters are required.

The input AFP print stream is read from the INFILE DD statement and the output AFP print stream is written to the OUTFILE DD statement. Look in member AFPCOPYX of JCLLIB to find an example of this utility.

On MVS systems, use `DD:INFILE` as the input file in the JCL. Use `DD:OUTFILE` as the output file in the JCL. Use `DD:BM,8205,8209` on the SysUT2 DD statement.

Example AFPCOPY /infile /outfile

AFPDUMP Use the AFPDUMP utility to create a text file from a print-ready AFP file. This utility produces a formatted dump of an AFP output file.

Program names

Windows AFPDPW32.EXE

Syntax

AFPDPW32 /I /H

Parameter Description

/I	The file name of the AFP file to dump. Include the extension. The output is named <i>filename.dds</i>
/H	(Optional) Include this parameter to tell the utility to also dump the HEX values.

The AFPDUMP utility reads a print-ready AFP file and creates a text file which contains English explanations of the AFP printer commands. If you include the /H option, the utility lists the hexadecimal characters for each AFP command before the its English explanation.

In addition to reading a print-ready AFP file, the AFPDUMP utility can also read AFP overlays (page segments) and AFP font files (coded font, codepage, and character set files).

You can use the AFPDUMP utility to check:

- Which fonts are being used in an AFP print ready or overlay.
- If text is correctly output in an AFP print ready or overlay.
- Which codepage and character set files are used by a coded font file.
- What characters and code points are defined in an AFP codepage file.
- If the characters named in the codepage file match those in the character set file.

Example

AFPDUMP /I=filename.pds /H

This converts the *filename.pds* file into the *filename.dds* file which you can read using any text editor.

NOTE If the file to be converted does not have a PDS extension, you must include the extension when you enter the command. If the file has no extension, add a period (.) after the file name.

Here is an example AFPDUMP output file (*name.dds*) with the /H option:

```

....5A 0010 D3A8A8 00 0000
000,Begin,Document,16
....4040404040404040

',
.... 5A 0010 D3ABCC 00 0001
001,      Map,Medium Map,16
.... D7D3E4D640404040
      ,PLUO
.... 5A 0010 D3A8AF 00 0002
002,      Begin,Page,16
.... 4040404040404040

',
....      5A 0010 D3A8C9 00 0003
003,      Begin,Active Env. Group,16
....      4040404040404040

',
....      5A 0017 D3A6AF 00 0004
004,      Descriptor,Page,23
....      0000096009600007F8000A500000000
      ,unit base 0 0,L-Units 2400 2400,Page Size x=2040 y=2640
....      5A 0019 D3A69B 00 0005
005,      Descriptor,Composed Text,25
....      00000960096007F80A5000000000000000

....      5A 0010 D3A9C9 00 0006
006,      End,Active Env. Group,16
....      4040404040404040

',
....      5A 0010 D3A89B 00 0007
007,      Begin,Composed Text,16
....      4040404040404040

....      5A 0012 D3EE9B 00 0008
008,      Data,Composed Text,18
....      2BD306F700002D0002F8

      Set Text Orientation,6,0000,2D00
      NOP,2,
....      5A 0010 D3A99B 00 0009
009,      End,Composed Text,16
....      4040404040404040

....      5A 0010 D3A89B 00 000A
010,      Begin,Composed Text,16
....      4040404040404040

....      5A 0012 D3EE9B 00 000B
011,      Data,Composed Text,18
....      2BD306F700002D0002F8

      Set Text Orientation,6,0000,2D00
      NOP,2,
....      5A 0010 D3A99B 00 000C
012,      End,Composed Text,16

```

```

....      4040404040404040

....      5A 0010 D3A89B 00 000D
013,      Begin,Composed Text,16
....      4040404040404040

....      5A 000C D3EE9B 00 000E
014,      Data,Composed Text,12
....      2BD302F8

NOP,2,
....      5A 0010 D3A99B 00 000F
015,      End,Composed Text,16
....      4040404040404040

....      5A 0010 D3A89B 00 0010
016,      Begin,Composed Text,16
....      4040404040404040

....      5A 000C D3EE9B 00 0011
017,      Data,Composed Text,12
....      2BD302F8

NOP,2,
....      5A 0010 D3A99B 00 0012
018,      End,Composed Text,16
....      4040404040404040

....      5A 0010 D3A8BB 00 0013
019,      Begin,Graphic,16
....      4040404040404040
,
....      5A 0010 D3A8C7 00 0014
020,      Begin,Obj. Env. Group,16
....      4040404040404040
,
....      5A 001C D3A66B 00 0015
021,      Descriptor,Object Area,28
....      034301084B0000009600960094C020007B1000A60
,Unit Base 0 0,L-Units 2400 2400,Object Area Size
x=1969 y=2656
....      5A 0020 D3AC6B 00 0016
022,      Position,Object Area,32
....      011800000000000000002D00000000000000002D0000
,Object Area Origin 0 0, Object Area Orientation
(0, 90), Object Content Origin 0 0
,Use the current coordinate system
....      5A 000D D3ABBB 00 0017
023,      Map,Graphic,13
....      0005030420
,Scale-to-fit
....      5A 0020 D3A6BB 00 0018
024,      Descriptor,Graphic,32
....      F6160000000009600960000080007FFF80007FFF00000000
,No absolute picture units ,Unit base 0 ,Window
Coordinates xLeft=-32768 xRight=32767 yBottom=-32768 yTop=32767

```

```

....      5A 0010 D3A9C7 00 0019
025,      End,Obj. Env. Group,16
....      4040404040404040
          ,
....      5A 0056 D3EEBB 00 001A
026,      Data,Graphic,86
....
700C1010101000000040000000000A0168C00A011807C10C1000100030003000500
070006000220830003000000000000C706000000000100C514000000001500400030
0000004500400060000000

          Begin Segment Introducer, 12
          Set Color, 01
          Begin Area, c0
          Set Color, 01
          Set Line Type, 07
          Line, Point (4096 4096), Point (12288 12288), Point
(20480 28672),
          End Area, 00
          Set Arc Parameters, Point (12288 12288), Point (0 0),
          Full Arc, Point (0 0), Scale 1.000000
          Fillet, Point (0 0), Point (5376 16384), Point (12288 0),
Point (17664 16384), Point (24576 0),

....      5A 0010 D3A9BB 00 001B
027,      End,Graphic,16
....      4040404040404040
          ,
....      5A 0010 D3A89B 00 001C
028,      Begin,Composed Text,16
....      4040404040404040

....      5A 000C D3EE9B 00 001D
029,      Data,Composed Text,12
....      2BD302F8

          NOP,2,
....      5A 0010 D3A99B 00 001E
030,      End,Composed Text,16
....      4040404040404040

.... 5A 0010 D3A9AF 00 001F
031,      End,Page,16
.... 4040404040404040
          ,
....5A 0010 D3A9A8 00 0020
032,End,Document,16
....4040404040404040

```

AFPFMDEF Use the AFPFMDEF utility to create an AFP form definition resource file from a DAT file (see the example below).

Program names

Windows	AFPFMDEW.EXE
MVS	See the Documaker Server Installation Guide

Syntax AFPFMDEW /I

Parameter Description

/I	The data file's first two characters must be 1
----	---

The AFP form definition resource file defines certain print attributes, such as paper size, orientation, and duplex settings. This utility takes a text file with a DAT extension and compiles it into an AFP form definition resource file which is installed on the AFP printer.

Example AFPFMDEW /I=f1fsi.dat

The output file will have the same name as the data file, with an FDF extension.

Here is an example of a form definition DAT file:

```
*****
*
* FIELD LAYOUT
*
*,Medium Map Name;Medium Map Id;X Origin;Y Origin;Paper
Size;Orientation;Copies;
*
*           Stacking;Tray;Flash;Duplex;Print Quality;
*
* Medium Map Name => up to 8 character long use A-Z 0-9 $ # @
*
* Medium Map Id => 1 to 127
*
* X Origin      => 0 to 32767
*
* Y Origin      => 0 to 32767
*
* Paper Size    => L           Letter (default)
*                E           Executive
*                G           Legal
*                A           A4
*
* Orientation => 01 - Portrait
*                02 - Landscape
*                03 - Portrait 90
*                04 - Landscape 90
*
* Copies        => 1 to 255
*
* Stacking      => Yes or No
*
```

```

* Tray          => T1 - Source Drawer #1
*               T2 - Source Drawer #2
*               T3 - Source Drawer #3
*               T4 - Manual Feed
*               T5 - Envelopes
*
* Flash         => Yes or No
*
* Duplex        => Simplex, Duplex, or Tumble duplex
*
* Print Quality => Default, Lowest, or Highest
*
*
*-----
* Medium Map Name convention
*-----
*
*   Position   Values   Description
*   =====   =====   =====
*           0
*           L       Landscape
*           P       Portrait (default)
*           1
*           E       Executive
*           G       Legal
*           A       A4
*           L       Letter (default)
*           2
*           M       Tray selection
*           M       Manual
*           F       Envelope feeder
*           L       Lower
*           U       Upper (default)
*           3
*           L       Duplex
*           L       Long binding
*           S       Short binding
*           O       Simplex
*           ?       Unknown (default)
*
*
*
*****
;B1;1;0;0;L;01;1;Y;T1;Y;S;L;
*----Landscape [Paper Size] Upper Simplex
;LLU0;60;0;0;L;02;255;Y;T1;Y;S;D;

```

A standard form definition file called **F1FMMS** is included with the system. It contains medium maps of all possible combinations of orientations, paper sizes, duplex mode and so on.

This form definition file should be resident on the printer. On MVS, place it in a PDS which is available to the PSF procedure. On a Novell print server, define it as a device option in the printer profile in PSF2.

ARCCNV Use the ARCCNV utility to convert old DOS archive files into the newer FAP file format.

Program names

Windows ARCCVW32.EXE

Syntax

ARCCVW32 /O /N /F /I /R

Parameter Description

/O	The old archive directory, where the old archive files are stored.
/N	The new archive directory, where the new archive files will be stored.
/F	The forms directory, where the corresponding FAP files are stored.
/I	(Optional) The file name for a single file conversion.
/R	(Optional) The file name restart point.

Convert all forms and then run this utility. The Windows archive subdirectory must be empty when you start the utility, otherwise you may lose data.

Example

Here's an example of how you could use this utility:

```
ARCCVW32 /O=\PPS\RESLIB\SAMPCO\ARC /N=\PPSWIN\MSTRRES\SAMPCO\ARC /
F=\PPSWIN\MSTRRES\SAMPCO\FORMS
```

After the conversion is complete, you must copy the APPIDX.DFD file to the new archive subdirectory. These conversion steps are necessary for every company library in the DOS PPS system.

Error messages

Here are explanations of possible error messages you may receive:

NA load image failure on 'IMAGENAME' at offset OFFSET

This message generally appears if:

- The FAP file for 'IMAGENAME' is missing
- You entered an incorrect forms subdirectory on the command line

To correct this error, do the following:

- 1 Remove all files from WINDOWS archive subdirectory.
- 2 Correct command line parameters or find the FAP file for IMAGENAME.
- 3 Start the ARCCNVW utility again.

Cannot load file 'FILENAME.DAT'

This message generally appears if the data file is corrupted in DOS PPS archive. To correct this error, do the following:

- 1 Remove all files from the Windows archive subdirectory.
- 2 Edit the FILENAME.DAT file using text editor and correct the corrupted data.

- 3 Rearchive the FILENAME.DAT file using the DOS command line utility, REPLARC. Here is the syntax for the REPLARC utility:

```
REPLARC.EXE <CARFILENAME> <FILENAME.DAT>
```

For example...

```
REPLARC.EXE ARCHIVE.CAR B225F307.DAT
```

- 4 Then, restart the archive conversion. If the problem reappears, see the next topic.

Checking ARCHIVE.CAR file for possible corrupted data files Run the DOS utility RESTARC.EXE and redirect the output into a file. For example...

```
RESTARC.EXE ARCHIVE.CAR /V > FILE.BAT
```

Edit the FILE.BAT file using a text editor. The file will look similar to this one:

```
DOS Archive Restore Program
Usage
  restarc <archive file name> [/Verbose]
Example
  restarc archive.car
File B1DA85BC.POL is OK
File B1DA85BC.DAT is OK
File B1DA872F.POL is OK
File B1DA872F.DAT is OK
File B1DC356D.POL is OK
File B1DC356D.DAT is OK
File B1E1289C.POL is OK
File B1E1289C.DAT is OK
Done
```

You have to create a batch file using this file. The batch file should look like this:

```
RETRIEVE.EXE ARCHIVE.CAR B1DA85BC.POL
RETRIEVE.EXE ARCHIVE.CAR B1DA85BC.DAT
RETRIEVE.EXE ARCHIVE.CAR B1DA872F.POL
RETRIEVE.EXE ARCHIVE.CAR B1DA872F.DAT
RETRIEVE.EXE ARCHIVE.CAR B1DC356D.POL
RETRIEVE.EXE ARCHIVE.CAR B1DC356D.DAT
RETRIEVE.EXE ARCHIVE.CAR B1E1289C.POL
RETRIEVE.EXE ARCHIVE.CAR B1E1289C.DAT
RETRIEVE.EXE ARCHIVE.CAR B1EA6DB4.POL
RETRIEVE.EXE ARCHIVE.CAR B1EA6DB4.DAT
```

Where the file names at the end of each line are the same names as those in the original FILE.BAT file. Run the batch file to unzip all DAT and POL files from the ARCHIVE.CAR file. Then, enter the following command:

```
GREP -l -v [\ -~] *.DAT
```

NOTE You must enter a space after the backslash (\)

This command displays a list of files which contain non-ASCII characters. These are files you must fix. See the information on the error message *cannot load file FILENAME.DAT* for information on what to do next.

After you fix all files, run the ARCCNV utility.

ARCFIX Use the ARCFIX utility to evaluate and, if necessary, attempt to repair archive files.

NOTE The ARCFIX utility should only be used by those who understand archive well because of the risk of data loss.

Make sure you *back up your archive files* before you run this utility. Support Services will run this utility for you if you encounter problems. For information on how to contact Support Services, see [Finding the Right Utility on page 2](#)

Program names

Windows ARCFXW32.EXE

Syntax ARCFXW32 /I /A /F

Parameter Description

/I	Enter the name of the CAR file.
/A	(Optional) Include this parameter to repair all files.
/F	(Optional) Include this parameter to fix the table of offsets.

Please note that:

- Index files must be in current directory.
- You must have the following files in the working directory:

APPIDX.DFD

???.CAR (for example. ARCHIVE.CAR)

The ARCFIX utility first checks to see that each record in the CAR file has a corresponding APPIDX record and then deletes any extra APPIDX records.

If you use the /F parameter, the ARCFIX utility rebuilds the table of offsets at the end of the CAR file.

ARCMERGE Use this utility to combine two archives that have the same DFD files for the archive index, catalog, and CAR file. This utility can take a secondary archive and merge it into the main archive.

NOTE This utility only works with xBase files. It does not work on SQL, Oracle, or DB2 databases.

Program names

Windows ARCMW32.EXE

Syntax

ARCMW32 /I=secondary-archive-directory /INI

Parameter Description

/I	If the secondary archive is a database, enter the name of the index (APPIDX) file. If the secondary archive is a file, enter the name and path of the file. You can omit the remaining parameters.
/INI	Enter the name of the INI file you want the utility to reference.

Run this utility from your master archive environment. The ARCMERGE utility uses the FSIUSER.INI and FSISYS.INI files, along with your resources. You can include these additional INI options in the FSIUSER.INI file to specify the names of the split archive files/tables.

```
< ArcRet >
MergeLog =
```

Option Description

MergeLog	The name of the log file. The default is <i>ARCMERGE.LOG</i> . The utility creates this file in the Data directory.
----------	---

Example

Assume that D:\REL10\MSTRRES\ARCB is the secondary archive, to be merged into the main archive. This is the command you would use if the secondary archive is stored in files rather than in a database:

```
ARCMW32 /I=d:\rel10\mstrres\arcb
```

If the archive is in a database archive, you would enter this command:

```
ARCMW32 /I=arcb/appidxs /A=arcb/archives /C=arcb/catalogs
```

Here is an example of the ARCMERGE.LOG file:

```
--- ArcMerge ---

Rows copied from <archives> to<archive>, Day Mon DD HH:MM:SS YYYY
C: Original ArcKey: New ArcKey
.
.
End of copy, Day Mon DD HH:MM:SS YYYY

Rows copied from<catalogs> to<Catalog>, Day Mon DD HH:MM:SS YYYY
C:Catalog ID
.
.
End of copy, Day Mon DD HH:MM:SS YYYY

Rows copied from<catalogs> to<Catalog>, Day Mon DD HH:MM:SS YYYY
C:FIELD1,FIELD2,FIELD3,original ARCKEY:updated ARCKEY
.
.
End of copy, Day Mon DD HH:MM:SS YYYY
```

See also [ARCSPLIT on page 39](#)

ARCRET Use this utility to retrieve records from your archives and produce files that can then be sent to *plug-in* functions to generate additional output. You can choose from these plug-in functions:

Plug-in	Description
PLGGenPrint	Produces a batch file the GenPrint program can use to produce printed form set output. Use it to reprint your archives.
PLGGenArc	Produces a batch file the GenArc program can use to archive transactions to a different archive. Use it to help migrate your archive to another archiving method.
PLGTest	Used to test the retrieval results from the archive. This function does nothing with the output of the ARCRET utility.

For example, you can use the PLGGenPrint plug-in function to take output from the ARCRET utility, produce a batch file, and then run the GenPrint program to produce printed form sets.

You can also use the PLGGenArc plug-in to migrate traditional LAN flat file archives to host or server based SQL database format archives, or as a part of an upgrade or conversion to a Documange repository.

Before starting any migration, you should...

- Back up your archives.
- Read all of the documentation concerning this utility.
- Run some tests and verify the results. Testing should help you:

Determine if archiving and retrieving from the new system is working properly.

Decide the optimal size of the batch sets.

Estimate how long the entire process will take.

After you verify the test results, change all systems that do archiving and retrieval to use the new system. Do not use the old system while you are migrating from it.

NOTE After you complete the migration, hang onto your original archives for awhile—just in case. Only after you are satisfied that the new system is working properly and that all of your data has been migrated, should you consider dropping the old archive and old archive system.

Program names

Windows ARCRET.EXE

Syntax

ARCRET /INI /R /N /S /P /BEF /OR /AFT /DAL /NC /DB /K /REV /LOAD
/RUNDATE

Parameter Description

/INI	Specifies the INI file to use. If you omit this option, the utility uses the FSIUSER.INI file as the default.
/R	Defines the sequential row or record number to begin processing. The default is one.
/N	Tells the utility to stop after retrieving this number of sequential rows or records. The default is to retrieve all rows.
/S	Sets the number of records to include in a set or batch. The default is one transaction per set.
/P	Tells the utility to pause between sets. The default is off.
/BEF	Selects rows/records with a RUNDATE before this date (YYYYMMDD). The date you specify is not included in the set.
/AFT	Selects rows/records with a RUNDATE after this date (YYYYMMDD). The date you specify is not included in the set.
/OR	Changes the date comparison to mean <i>before or after</i> as opposed to the default <i>before and after</i> .
/DAL	Specifies a script file the utility should run to determine if a row/record is included in a set.
/NC	Tells the utility not to compile the DAL script. DAL scripts are compiled by default to improve performance.
/DB	Dumps debug information as the script runs. When you include this parameter, use a small set (/S) size.
/K	Keeps all intermediate files by renaming them instead of deleting them. Using this option can leave numerous files that you will have to remove manually.
/REV	Tells the utility to process the archives in reverse order. The last record is read first and the first record is read last. This only applies to xBase databases.

Parameter Description

/LOAD	This parameter lets you filter archive records by examining the fields in the loaded form set using a DAL script. You can update the ARCRET record members passed to DAL as the script executes. The /LOAD parameter names a script the system will execute after the form set is retrieved from archive and loaded. DAL had the ability to query form set field values as well as other form set members. Any changes the script makes to the loaded document are temporary, as the actual archived document cannot be changed. However, the script can change ARCRET member values, which represent the archive index record used to identify the document. By allowing DAL to update the ARCRET record members, the script can change the current record values that will be written to the output files generated by ARCRET. This may later influence any plug-in functions called via ARCRET.
/RUNDATE	Lets the user to specify a field to use instead of RUNDATE. The format of the field you designate should be YYYYMMDD.
/DATEFMT	Identifies the date format used within the RUNDATE field. If you omit this parameter, the utility assumes the RUNDATE field is stored in the typical YYYYMMDD format. The DATEFMT must be specified using a standard internal FAP date format. Additionally you can use the format X to indicate that the RUNDATE field is stored using the Skywire Software Hex Time format. If a PPS user has an archive that doesn't specify a RUNDATE field but instead uses the WIP CreateTime as the date field, it is usually stored in a Hex Time format. To do a retrieval from this archive, you would specify a command line with these parameters: <pre>ARCRET /RUNDATE=CreateTime /DATEFMT=X</pre> The /RUNDATE parameter is an existing parameter that identifies an alternative field to use as the RUNDATE. The /DATEFMT parameter indicates how to interpret the date information in the specified RUNDATE field.

Transaction sets are passed to the plug-in functions you define in the INI file for additional processing. If no plug-in functions are defined, nothing happens to the retrieved records. All temporary files created during the run are deleted when the utility stops.

INI files By default, the utility loads the FSIUSER.INI and the INI file defined by this option in the FSIUSER.INI file:

```
< Environment >
    FSI SYSINI =
```

To specify a different INI file, use the INI parameter as shown here:

```
ARCRET /INI=My.INI
```

Defining plug-in functions Define the plug-in functions in the ArcRet control group as shown here:

```
< ArcRet >
    PlugInMod = NAME.DLL
    PlugInFunc = Function
    PlugInFunc = Function2
    PlugInFunc = Function3
    PlugInFunc = Function4
    PlugInFunc = Function5
```

The PlugInMod option defines the DLL file that contains the functions you want to use. The PlugInFunc option defines the function name of a plug-in compatible function.

You can define up to five plug-in functions which will be executed in the order they are defined in the INI file. If no functions are defined, the utility displays this message:

```
Warning: No plug-ins loaded.
```

NOTE The ARCRET utility will continue to run even if no plug-in functions are defined.

Plug-in functions must conform to a specific prototype and will be passed specific information about the files to process.

You can also use the DLL->FunctionName method for naming plug-ins. This lets you keep plug-ins in multiple DLL files. Here is an example:

```
PlugInFunc = DLL->FunctionName
```

If the option does not contain the “->” to indicate a DLL name is specified, the system assumes it should use the PlugInMod option to locate the DLL for this function.

Archive index field mapping

The ARCRET utility is designed to find all the relevant information about your archive setup under the ArcRet control group. In general, the INI options required by the ARCRET utility are almost identical to those used by the GenArc program.

With a few exceptions, the settings referenced are the almost the same as those used and required by the AFEMAIN program. One exception is the Trigger2Archive control group.

The ARCRET utility uses this Trigger2Archive control group to re-create the NEWTRN file. If you use the GenArc program to produce your archives, you already have this group to map the fields from the NEWTRN file to the APPIDX (application index) file.

Since the AFEMAIN program does not use this control group, you may have to add it to your INI file. Here is an example of what you would need to add:

```
< Trigger2Archive >
    ArcField = TrnField
    ArcField2 = TrnField2
    ...
```

Each *Trn* field is defined in the TRNDFDFL.DFD file which is specified using the TrnDFDFile option in the Data control group. Each *Act* field is defined in the APPIDX.DFD file which is specified using the AppIdxDFD option in the ArcRet control group.

If you do not have the Trigger2Archive control group defined, the ARCRET utility tries to match the fields in the TRNDFDFL.DFD and APPIDX.DFD files by name. If no field names match, an error message appears and the ARCRET utility stops.

Skipping rows/records

By default, the ARCRET utility selects each archive index record in sequence. Since you may not always want this, several parameters are included so you can designate which rows/records you want to process.

Keep in mind that the ARCRET utility presumes every index record is a potential candidate for selection.

The first record in the first index file is read and checked against the selection criteria (unless you included the /REV parameter). The next record is then read and so on until all the records in the index have been examined. Each record is either accepted or rejected based on the parameters you specify.

You can use the /R parameter to skip a specific number of sequential transactions.

You can also use the /N parameter to indicate a maximum number of ~~accepted~~ transactions you want to process. By default, the ARCRET utility continues until the last index row/record is processed.

You use the /R and /N parameters to control where and when to stop processing. The /REV parameter lets you specify whether the utility should start at the beginning or the end of the file. These parameters can be useful when you have a large number of rows/records that will take a long time to process.

For example, suppose you can only process 1000 transactions a day with the plug-in you want to use and you have 3000 transactions in your archive. Potentially, that means it would take three days to process these transactions using your plug-in.

Assuming you are not using any other parameters, you would enter these commands:

```
Day 1: ARCRET /N=1000
Day 2: ARCRET /N=1000 /R=1000
Day 3: ARCRET /R=2000
```

Notice, that the first day, you did not have to specify the /R command, but you did specify to stop after processing 1000 transactions. The second and third days, you did specify how many leading transactions to skip. Note that on the final day, the /N parameter was omitted. If you know the remaining set of records will not exceed your maximum, you can omit this parameter.

Controlling the number of transactions sent to the plug-ins

Each time a set of matching transactions are located and retrieved, the ARCRET utility calls the plug-in functions to process those transactions. By default the ARCRET utility searches until a single matching transaction is found, retrieves the associated form set, and then calls the plug-ins. Therefore, the default set size is one transaction.

For some plug-in functions, a larger set size will improve performance—especially if the plug-in has excessive startup or shutdown time requirements. Use the /S parameter to designate the number of rows/records to include in a set before the ARCRET utility calls the plug-in functions.

NOTE If there are not enough matching transactions found in the index file, the plug-ins are called with however many matching transactions were found. If you create a plug-in, keep in mind the plug-in should make no assumptions about the number of transactions in the sets.

You can also use the /P parameter with sets. This parameter tells the utility to pause as each set is processed and wait until you press **ENTER** before building the next set. This parameter is useful if you need to examine or copy the output files produced by the plug-ins before starting the next set.

Selecting records by date

There are several parameters you can use to select transactions which fall within or outside a given date range. When specifying dates for these parameters, be sure to use the YYYYMMDD format.

Keep in mind that these parameters assume the RUNDATE or CREATETIME variables are the names of the transaction date fields. The utility first looks for RUNDATE. If it is not found, the utility looks for CREATETIME (the name used by standard AFEMAIN archives).

If the utility finds RUNDATE it looks for the record which should be in YYYYMMDD format. When using CREATETIME, the utility assumes the date is in the internal HEXTIME format and converts it to YYYYMMDD format, comparing it to the dates specified. If neither field is found, the ARCRET utility displays an error and stops processing.

NOTE For the remainder of this topic, assume that RUNDATE means either RUNDATE or CREATETIME and that the data value will be in YYYYMMDD format.

Use the /BEF parameter to tell the utility to select records with a RUNDATE value that falls *before* a given date. Transactions with the specified date are excluded. For instance, to select transactions archived before 2001, you would specify:

```
/BEF=20010101
```

No records after December 31, 2000 are selected.

To select transactions with a RUNDATE that falls *after* a given date, use the /AFT parameter. Transactions with the specified date are excluded. For instance, to select the transactions archived in 2001, specify:

```
/AFT=20001231
```

All records after December 31, 2000 are selected.

You can also use these parameters together to specify a range. When you use both parameters, the utility assumes you want a logical AND comparison, so the /BEF date should fall after the /AFT date. For example, to select all records for the year 2000, you would specify these parameters:

```
ARCRET /AFT=19991231 /BEF=20010101
```

This tells the utility to select all transaction with a RUNDATE that falls within the two dates. You can also include the /OR parameter to omit records that fall within a certain range.

When you use the /BEF, /AFT, and /OR parameters, the utility excludes the transactions which fall within the dates specified. For example, to select all records except those which fall in the year 2000, you would specify these parameters:

```
ARCRET /BEF=20000101 /OR /AFT=20011231
```

Note that when you use the /OR parameter, you identify the earlier date using the /BEF parameter and the later date with the /AFT parameter.

NOTE Where you place the parameters in the command does not matter.

The ARCRET utility tries to validate the date ranges you specify based on the parameters you enter. If the utility detects a combination of parameters or values that do not make sense, it displays an error message and stops.

Using DAL to select records

You can also include a DAL script to provide the final approval or rejection of a particular transaction. Note that the utility processes all other parameters before it executes the DAL script.

The /DAL parameter names the DAL script you want to execute on each transaction. Before calling the script, the APPIDX record variables are converted into DAL variables using the standard DAL DB naming convention.

The DAL DB nomenclature associates all the row/record variable names with a table name. The table name is typically specified in the script or is the name of the table being referenced. Because DAL does not actually open the database and because there may be more than one index file, all record members are associated with the name *ARCRET*.

For instance, suppose your APPIDX has these members specified in the DFD file:

```
< Fields >
  FieldName = Key1
  FieldName = Key2
  FieldName = PolicyNum
  FieldName = RunDate
  FieldName = ArcKey
  FieldName = FormsetId
```

These fields would be referenced using the following names in any script associated with the ARCRET utility.

```
ArcRet.Key1
ArcRet.Key2
ArcRet.PolicyNum
ArcRet.RunDate
ArcRet.ArcKey
ArcRet.FormsetId
```

Note that since DAL only supports STRING, LONG, and DECIMAL data types, some variables may be converted to the closest matching type. When in doubt, assume the data will be of the STRING type for comparison purposes.

Since DAL has no knowledge of how a field will be used, always use the FORMAT and DEFORMAT functions where appropriate in your DAL scripts. For DFD members that hold DATE values, be sure to use the appropriate DATE functions, such as DATE2DATE, to convert values to a standard format before comparing them.

For example, if you want to select all transactions where the `key2` variable contains `TEXAS`, you would write a DAL script similar to this one:

```
IF ArcRet.Key2 = "TEXAS"
    RETURN("Yes");
END
RETURN("NO");
```

NOTE The return value from the script is important. If the script returns `Yes`, that means to include the transaction in the set. Any other return value – including omitting a return value—excludes the transaction from the set. The case of the word `Yes` in the returned value is not important.

If you saved this script using the name `MATCH.DAL`, you would enter this command:

```
ARCRET /DAL=MATCH.DAL
```

For performance reasons, the utility normally pre-compiles your DAL scripts in memory before executing them. This makes each subsequent execution of the script faster than if the script was not pre-compiled. You can, however, turn off this behavior by including the `/NC` parameter. Typically, you would only include this parameter for debugging purposes.

You can also use the `/DB` parameter to debug scripts. This parameter produces lots of output, so be sure to send the console output to a file using the `<">filename"` parameter on the command line. Also, use the `/S` parameter to limit the run to only a few transactions. Otherwise, the amount of output will be overwhelming.

Keeping intermediate files

Normally, the utility removes intermediate files after each batch set is complete and before it starts the next set. Use the `/K` parameter to keep these files for each batch sets. Each set of files is renamed with an extension that matches the set (batch) count. For instance, the files from the first batch of records will be renamed as `*.1`; the second set of files as `*.2`; and so on.

Be careful using this parameter. Depending upon the size of each transaction and the number of transactions retrieved, using this parameter can consume a lot of file space. Keep in mind you must remove these files when the ARCRET utility finishes.

Batch queuing

Normally, the ARCRET utility retrieves transactions interactively while sequentially accessing your archive index systems. In some instances, however, the underlying archive system does not support retrieving transactions while sequentially processing the records.

This is the case with Documanage. Since Documanage serves as both the archive index and storage system, it cannot currently be accessed in the same manner as a database. Therefore, use the `/BQ` option to queue sequential transactions in memory before attempting to retrieve the associated files.

Queuing the transactions this way consumes more memory than would otherwise be required. In general, you can determine the index record size by examining the `APPIDX.DFD` file you use. Add some extra for overhead and then divide that into the maximum amount of memory that you wish to consume—over and above that used by the ARCRET utility itself. To determine this, use the Windows NT Task Manager to get an idea of how much memory is in use while you run the utility on a single transaction.

Batch queuing also affects performance because the utility has to reset the sequential index search between each set of records. For instance, suppose it reads and queues 100 transactions in a set. After the set is processed, it has to re-read sequentially from the beginning of the index back down to the 100th record to prepare for the next set. For each set of transactions, this *resetting* takes longer and longer, because it has to start at the beginning of the index each time.

NOTE Only use the /BQ parameter when the source archive system that ARCRET utility is reading does not support the interactive retrieval of documents while sequential reading the index records.

Using the PLGTest Plug-in

This plug-in prints a few messages each time it is called. Use it for testing purposes only. For instance, you would use this plug-in before using one of the other plug-ins to make sure you are successfully retrieving records and that those records match your criteria.

You can include the /P (pause) option to temporarily stop processing between batches. This lets you examine the intermediate files.

To use this plug-in, make sure you have these options in your INI file:

```
< ArcRet >
  PlugInMod = PLGW32.DLL
  PlugInFunc= PLGTest
```

Using the PLGGenPrint Plug-in

This plug-in accepts output from the ARCRET utility and executes the GenPrint program to produce printed output of archived transactions.

NOTE Before you use this plug-in, make sure you have a batch setup that can run the GenPrint program to produce the printed output you want.

To specify this plug-in, include these INI settings:

```
< ArcRet >
  PlugInMod = PLGW32.DLL
  PlugInFunc = PLGGenPrint
```

Note that case's important when specifying the name of the plug-in. You must specify it *exactly* as shown here.

The ARCRET utility produces an NA/POL file set similar to that produced by the GenData program. This includes a NewTrn file that is created from the archive index using the options in the Trigger2Archive control group. The plug-in then converts the NewTrn file into a batch (RCBDFDFL.DFD) compatible file and starts the GenPrint program to complete the process.

The conversion from the NewTrn type output of the ARCRET utility to a recipient batch file is accomplished by matching the field names between the two files. No addition INI options are necessary. The utility simply uses the TRNDFDFL.DFD file and the RCBDFDFL.DFD file (or whatever names you give these files in your INI settings) and match the fields by name. The utility automatically handles any other necessary conversions.

The plug-in produces a recipient batch record for each recipient found in the form set associated with the transaction. Because only one batch file is produced, all recipients are written to the same file.

Once the batch file is produced, the utility internally re-maps the INI options for the following items, writes a temporary INI file, and then executes the GenPrint program specifying that INI file to produce the batch output.

```
< Environment >
    FSISYSINI =
< Print_Batches >
    ...
< Exclude_Batches >
    ...
< Data >
    NAFile      =
    POLFile     =
    NewTrn      =
```

Option Changes

Environment control group

FSISYSINI This option is eliminated. All of the current options from both INI files were placed in memory while executing the ARCRET utility, so the file is not necessary in the subsequent run of the GenPrint program.

Print_Batches control group

all options All batches defined under this group are eliminated except for the first alphabetical group. The utility uses the first option as the batch to be processed. Make sure the printer settings for this batch are appropriate. The utility changes the name of the BCH file associated with this option to match the file produced by the plug-in.

Exclude_Batches control group

all options The utility eliminates all of the options in the control group to make sure the one batch left intact is processed.

Data control group

NAFile	This option is changed to match the output from the ARCRET utility.
POLFile	This option is changed to match the output from the ARCRET utility.
NewTrn	This option is changed to match the output from the ARCRET utility.

In general, all the temporary files that are produced have the same (8-digit hexadecimal) name and a different extension. This includes the temporary INI file the utility produces for the GenPrint program.

If GenPrint errors occur, remember the output of the GenPrint program is written to the error file specified in the Data control group and is not controlled by the plug-in or the ARCRET utility.

Using the PLGGenArc Plug-in

This plug-in accepts output from the ARCRET utility and executes the GenArc program to archive your transactions to another system.

NOTE For this discussion *source* refers to archive system where the ARCRET utility is reading transactions. *Destination* refers to the new archive system where the GenArc program will save transactions retrieved from *source*.

Before you use this plug-in, make sure your system can successfully...

- Retrieve transactions from your *source* archive system. This involves INI options in a file which this discussion refers to as the *FSIOLD.INI* file.
- Add transactions to your *destination* archive system. This involves INI options in a file which this discussion refers to as the *FSINEW.INI* file.

Make sure the TriggerToArchive control group is in the (source) FSIOLD.INI file and that you map the archive index fields defined in the APPIDX.DFD file to the corresponding fields in the TRNDFDFL.DFD file. The field names between the two DFD files don't have to be the same. In the FSIOLD.INI, add these options:

```
< ArcRet >
  PlugInMod = PLGW32.DLL
  PlugInFunc = PLGGenArc
< GenArcPlugIn >
  INIFile    = FSINEW.INI
```

The options in the ArcRet control group identify the plug-in function the ARCRET utility will use. The INIFile option in the GenArcPlugIn control group identifies the INI file that should be supplied to the GenArc program by the plug-in.

Use a command similar to this one to run the ARCRET utility:

```
ARCRET /INI=FSIOLD.INI
```

This tells the ARCRET utility to use the correct INI file to retrieve from your source archive. When the plug-in is called, the batch files created by the ARCRET utility are automatically added to the INI file you named. In part, the changes will be to several options in the Data control group, such as those shown here:

```
< Data >
  NAFile =
  NewTrn =
  POLFile =
```

The plug-in then starts a GenArc session and sends to it a temporary INI file created by combining these options. If you have successfully tested your GenArc program setup using the INI file you named, it should work without further intervention.

EXAMPLES

These examples begin with the base FSIUSER.INI file in RPEX1 and use the standard xBase and CARFile archives. Here is an example of the ArcRet control group:

```
< ArcRet >
  AppIdx      = ARC\AppIdx
  AppIdxDfd   = DefLib\AppIdx.Dfd
  ARCPATH     = [CONFIG:Batch Processing] ARCPATH =
  Arrangement= Stack
  CARFile     = ARCHIVE
  CARPath     = [CONFIG:Batch Processing] CARPath =
  Catalog     = ARC\Catalog
  ExactMatch  = No
  Key1        = Company
  Key2        = Lob
  KeyID       = PolicyNum
  LBLimit     = 500
  PlugInFunc  = PLGGenArc
  PlugInMod   = PLGW32
  TempIDX     = ARC\Temp
```

Only those changes required to get each archive system to work are shown.

ODBC/SQL server changes

Make these changes. Options not listed remain the same as those in the standard ArcRet control group from RPEX1. For SQL, you must change the CARData column from the default VARCHAR to a BLOB. You must also have a special restart table DFD to change the LASTRECORD column VARCHAR length.

```
< DBHandler:ODBC >
  CreateTable= Yes
  CreateIndex= No
  InstallMod  = sqw32
  InstallFunc= SQInstallHandler
  UserID     = sa
  Passwd     =
  Server     = MS SQL FSINTSRV07
< DBTable:CATALOG >
  DBHandler = ODBC
  UniqueTag = catalogID
< DBTable:APPIDX >
  DBHandler = ODBC
< DBTable:ARCHIVE >
  DBHandler = ODBC
  UniqueTag = arckey
< DBTable:RESTART >
  DBHandler = ODBC
< Archival >
  ArchiveMem = Yes
< ArcRet >
  RestartTable= Restart
  CARFileDFD  = carfile.dfd
  RestartDFD  = restart.dfd
```

DB2 changes Make these changes. Options not listed remain the same as those in the standard ArcRet control group from RPEX1.

```
< DBHandler:DB2 >
  BindFile = ..\dll\db2lib.bnd
  CreateTable= Yes
  CreateIndex= No
  Database = ARCDL
  UserID    = user
  Passwd    = admin
< DBTable:CATALOG >
  DBHandler = DB2
< DBTable:APPIDX >
  DBHandler = DB2
< DBTable:ARCHIVE >
  DBHandler = DB2
< DBTable:RESTART >
  DBHandler = DB2
< Archival >
  ArchiveMem = Yes
< ArcRet >
  RestartTable= Restart
```

Documange changes Make these changes. Options not listed remain the same as those in the standard ArcRet control group from RPEX1.

```
< Archival >
  ArchiveMem    = Yes
  UseRestartTable = No
< DBHandler:PO >
  Cabinet       = RPEX1
  Domain        = docucorp
  Password      = password
  UserID        = user
< DBTable:APPIDX >
  DBHandler     = PO
< DBTable:ARCHIVE >
  DBHandler     = PO
< PO:RPEX1 >
  FileType      = DAP
  FolderBy      = Company,Lob,PolicyNum
  NameDocBy     = ARKEY
```

ARCSPLIT Use the ARCSPLIT utility to back up all or part of your archives or split archive data based on a cut-off date or on a DAL script. Use the ARCMERGE utility to combine an archive split with this utility.

NOTE This utility only works with xBase files. It does not work on SQL, Oracle, or DB2 databases.

You can use the following parameters or INI options or a combination of both to control how this utility splits archives.

Program names

Windows ARCSPLIT.EXE

Syntax

ARCSPLIT /D /DAL /ED /INI /L /LOG /N /NCF /NSCF /NSIF /O /P /R /SD /SET /SRCH

Parameter Description

/D	(Optional) Include this parameter if you want the utility to delete any existing split archive files before it creates new ones.
/DAL	(Optional) Enter the name of the DAL script you want to use. This parameter overrides the DALScript and RunDALScript INI options.
/ED	Enter the date after which to end the search, in YYYYMMDD format. This parameter overrides the SplitToDate INI option.
/INI	Enter the name of the INI file you want the utility to reference. If you are using the default INI file (FSIUSER.INI) and it is in the current directory, you do not have to specify it.
/L	(Optional) Enter the size limit for the new CAR files the utility creates. For instance, if you want to set the maximum size at 100KB, enter 100. If you want to set the limit at 1,400,000KB, enter 1400. This parameter overrides the EnableCARFileSize and CARFileSize options.
/LOG	(Optional) Enter the name of the file into which the utility should write error messages. This parameter overrides the LogFile INI option.
/N	(Optional) Enter the number of records you want to process. This parameter overrides the RecordsToProcess INI option.
/NCF	Enter the full name you want to assign to the new catalog file. The utility defaults to the current MRL path. This parameter overrides the SplitCatalog INI option.
/NSCF	Enter the full name of the newly split archive (CAR) file. The utility defaults to the current MRL path. This parameter overrides the SplitCARFile INI option.

Parameter Description

/NSIF	Enter the full name of the newly split index (IDX) file. The utility defaults to the current MRL path. This parameter overrides the SplitAppIdx INI option.
/O	(Optional) Use this parameter to override the required date range when splitting an archive
/P	(Optional) Purge from the master archive the records which are split. If you omit this parameter, the specified records are written to the directory defined in the INI options and the master archive is not affected. This parameter overrides the PurgeRecords INI option.
/R	Enter the number or records to skip before processing occurs. The default is zero which tells the utility to start with the first record. This parameter overrides the RecordsToSkip INI option.
/SD	Enter the date on which to start the search, in YYYYMMDD format. This parameter overrides the SplitFromDate INI option.
/SET	Use to specify which configuration settings to use. You can have multiple configuration settings in the INI file. Keep in mind that command line parameters override INI settings, except/SET.
/SRCH	Enter <i>RunDate</i> to search the records based on the RunDate. Enter <i>ArchivedDate</i> to search the records based on the date on which they were archived. This parameter overrides the SearchDateBy INI option.

This utility uses the current master resource library (MRL) as the base for splitting the archive. If the archive you intend to split is not part of the current MRL, you will get incorrect output.

INI options

In addition to parameters, you can also set up INI options to use this utility. The utility looks in the INI file defined by the /INI parameter for these options. If you omit the /INI parameter, the utility looks in the FSIUSER.INI file.

These INI options are required. These options tell the ARCSPLIT utility where to find the source archive files you want to split.

```
< ArcRet >
  AppIdx = arc\appidx.dbf
  ArcPath = arc\
  CARFile = archive
  CARPath = arc\
  Catalog = arc\catalog
```

The utility also looks for options that specify the names of the split archive files and provide other information. These options are located in the ArcSplit and ArcSplitConfig control groups:

```
< ArcSplit >
  ArcSplitConfig =
  DefConfig =
< ArcSplitConfig:TEST1 >
  SplitDays =
```

```

RunDALScript      =
DALScript         =
RecordsToProcess  =
RecordsToSkip     =
SearchDateBy     =
SplitAppIdx       =
SplitCARFile      =
SplitCatalog      =
SplitFromDate     =
SplitToDate       =
PurgeRecords      =
CARFileSize       =
EnableCARFileSize =
AllowChanges      =
LogFile           =

```

Option	Description
ArcSplit	
ArcSplitConfig	Lets you assign a name to your archive split settings. You can create as many ArcSplitConfig options as you need. Each group of settings represents a control group with individual options. Below is an example of a setting named <i>TEST1</i>
DefConfig	For the ARCSPLIT utility, if you define the DefConfig option and omit the /SET parameter, the utility uses the DefConfig option to look for the ARCSPLIT configuration settings. If you set up multiple ArcSplitConfig options for Documaker Workstation, you can use this option to designate a default.
ArcSplitConfigTEST1	
SplitDays	(Optional) This option is not used by the ARCSPLIT utility, but is used by Documaker Workstation. Enter the number of days you want the utility to add to the start date to determine the end date. The utility then splits the archive between the start and end dates.
RunDALScript	(Optional) Enter Yes if you want the utility to run the DAL script you specified with the DALScript option. The /DAL parameter overrides this option.
DALScript	(Optional) Enter the name of the DAL script you want to run. Use the RunDALScript option to tell the utility if it should run the DAL script you specify with this option. The /DAL parameter overrides this option.
RecordsToProcess	(Optional) Enter the number of records to process. Use this option if you have a very large archive and you want to limit the number of records processed at one time. The /N parameter overrides this option.
RecordsToSkip	(Optional) Enter the number of the record the utility to process first. If you omit this option, the utility starts with the first record. The /R parameter overrides this option.

Option	Description
SearchDateBy	(Optional) Enter <i>RunDate</i> to search the records based on the RunDate. Enter <i>ArchivedDate</i> to search the records based on the date on which they were archived. The /SRCH parameter overrides this option.
SplitAppIdx	(Optional) Enter a name for the newly split IDX file, such as APPIDX1. The utility stores the data in the file you specify. If the file exists and you have set the /D parameter, the utility overwrites the file. If you have not set the /D parameter and the file exists, the utility stops. The /NSIF parameter overrides this option.
SplitCARFile	(Optional) Enter a name for the newly split CAR file, such as ARCHIVE1. The utility stores the data in the file you specify. If the file exists and you have set the /D parameter, the utility overwrites the file. If you have not set the /D parameter and the file exists, the utility stops. The /NSCF parameter overrides this option.
SplitCatalog	(Optional) Enter a name for the newly split catalog file, such as CATALOG1. The utility stores the data in the file you specify. If the file exists and you have set the /D parameter, the utility overwrites the file. If you have not set the /D parameter and the file exists, the utility stops. The /NCF parameter overrides this option.
SplitFromDate	(Optional) Enter the date on which you want the split to begin. The default is the current date. The format is MM/DD/YYYY. The /SD parameter overrides this option.
SplitToDate	(Optional) Enter the date on which you want the split to end. The default is the current date. The format is MM/DD/YYYY. The /ED parameter overrides this option.
PurgeRecords	(Optional) Enter Yes if you want the utility to purge from the master archive the records it split from the archive. The default is No, which tells the utility to copy but not delete those records. The utility stores the copied records in the files and directories you specified. The /P parameter overrides this option.
CARFileSize	Enter a number between one (1) to 14,000 to define the size for the CAR file. If you enter one (1), the utility interprets that as 100,000 bytes (100 KB). If you enter 14,000, the utility interprets that as 1,400,000,000 bytes (1,400,000 KB). Omit this option if the EnableCARFileSize option is set to No.

Option	Description
EnableCARFileSize	(Optional) This option turns on and off the related radio button field on the ArcSplit window in Documaker Workstation and also tells both Documaker Workstation and the ARCSPLIT utility whether to use the CARFileSize option. The default is No. The /L parameter overrides this option.
AllowChanges	(Optional) Lets you change settings from a window. The default is No.
LogFile	(Optional) Enter the name of the file into which the utility should write any error messages. Here is an example: <pre>LogFile = c:\errlog.txt</pre> The /LOG parameter overrides this option.

Using ARCSPLIT with DAL scripts

You can use DAL scripts to when you split or back up an archive. The ARCSPLIT utility makes available to the DAL script all of the APPIDX column values you specify. This script can only return Yes or No. If anything else is returned, the system defaults to No.

All field names specified in the script file must include the word **ARCSPLIT**, such as **ARCSPLIT.KEY1**. This is required in case multiple index files are in use.

NOTE Refer to the [DAL Reference](#) for information on the DAL functions you can use to create the scripts.

Assume the following form sets are stored in the archive, the INI options are set as shown below, and the DAL scripts COMPANY.DAL and COMBINED.DAL exist in the DefLib directory.

Table 1:

KeyID	Key1	Key2	Date archived
AA	Sampco	LB1	03/01/1999
BB	Sampco	LB2	03/01/1999
CC	FSI	GL	03/02/1999
DD	FSI	GF	03/02/1999
EE	MyCompany	GO	03/03/1999

Also assume these INI options are set:

```
< ArcRet >
SplitAppIdx    = arc1\AppIdx1.dbf
SplitCARFile   = arc1\Archive1.car
SplitCatalog   = arc1\Catalog1.
```

The COMPANY.DAL script looks like this:

```
If ARCSPLIT.Key1 = "FSI" then Return ("Yes");
                        Else Return ("No");
End
```

The COMBINED.DAL script looks like this:

```
if (ARCSPLIT.KEY1 = "FSI " AND ARCSPLIT.KEY2 = "GL ") then
Return ("YES");
Else
Return("NO");
End
```

NOTE Make sure the value you specify matches the field length defined in the DFD (Database Field Definition) file. In this example, the field length of KEY1 is four characters and the search value should be "FSI " (with a space between I and the ending quotation mark) instead of "FSI".

Based on these assumptions, this table shows the results if you enter the following commands to run the ARCSPLIT utility:

If you enter...	The result is..
ARCSPLIT /sd=19990301 / ed=19990301 / ini=fsiuser.ini	Records AA and BB are written to the archive files (ARCHIVE.CAR, APPIDX1.DBF, CATALOG1.DBF, APPIDX1.MDX, and CATALOG1.MDX) in the arc1 directory. The records in the master archive are not changed.
ARCSPLIT /sd=19990301 / ed=19990301 / ini=fsiuser.ini	Records AA and BB are written to the archive files in the arc1 directory. The records in the master archive are not changed. The APPIDX1 and CATALOG1 files will be flat files.
ARCSPLIT /sd=19990301 / ed=19990301 / ini=fsiuser.ini /p	Records AA and BB are written to the archive files in the arc1 directory. These records are also deleted from the master archive.
ARCSPLIT /dal=deflib\company.dal ini=fsiuser.ini	Records CC and DD are written to the archive files in the arc1 directory. The records in the master archive are not changed.
ARCSPLIT /dal=deflib\combined.dal ini=fsiuser.ini	Record CC is written to the archive files in the arc1 directory. The records in the master archive are not changed.

See also [ARCMERGE on page 24](#)

ARCVIEW You can use the ARCVIEW utility to view Documaker archive files checked into the Documanager archive system. This utility only runs under 32-bit Windows.

Program names

Windows	ARCVWW32.EXE
---------	--------------

You do not run this utility from the command line. Instead, you simply register this utility as the program you want to use to view Documanager files. To use this utility, follow these steps:

- 1 Register the Documanager file extension (DPA) in Windows so the operating system will automatically use the ARCVIEW utility to view these files.
- 2 Set the environment variable FSIPATH to point to the directory where the INI file for the AFEMAIN program is stored. Here is an example:

```
FSIPath = d:\rpexl
```

NOTE The AFEMAIN program is the executable file for Documaker Workstation.

- 3 Place a menu file, similar to the MEN.RES file used by Documaker Workstation, in the directory specified by the FSIPath option. The name of the menu file should be *ARCVIEW.RES*

NOTE You can edit this file to remove functionality you do not want to include.

- 4 Edit the FILETYPES.INI file on the computer where the Documanager server runs. Add the DPA file extension to the list of file types to view with the ARCVIEW.EXE program. This causes the Documanager client to use the viewer registered in Windows instead of the default Documanager viewer.

You can now click on Documaker archive files in Windows Explorer to display them.

ATPHDR

Use the ATPHDR utility to add missing header information for Xerox fonts if you experience this problem when you insert Xerox fonts into a font cross-reference file (FXR) using the Font Manager.

NOTE Docuview, which is part of Docusave workstation, requires Xerox resources to be padded to fill 512 byte blocks. Some old Xerox resources built with prior versions of the system do not meet this criteria. You can use this utility to read and update those Xerox resources so they can be used by Docuview.

Program names

Windows ATPHDRW.EXE

Syntax

ATPHDRW /I

Parameter Description

/I	Enter the input file name. Omit the extension.
----	--

The input file must have the extension *TMP*. The utility will create an output file with the extension *FNT*.

BARR2MVS Use the BARR2MVS utility when you have a Xerox Metacode spool file which is one long record and you want to separate it into separate records for MVS.

NOTE This program is only available on MVS.

Metacode print spools which are created on PCs and which use JES2 format, contain carriage return/line feeds (CR/LF) at the end of each record. However, some Metacode print spools created on a PC may contain binary data which happens to contain a carriage return/line feed. In this case, uploading the spool file with CR/LF translation will produce an invalid Metacode print spool file on MVS.

In this case, create the Metacode print spool using the BARR format, upload the original Metacode print spool file as binary, with no CR/LFs, and use the BARR2MVS utility to convert the BARR formatted print spool file into a Metacode print spool file with separate records on MVS.

Example Here is an example to show you how to do this:

First create a PS (Physical Sequential) file on MVS with DCB=U,0,23200. Name this file *WINFILE*. Then, create another PS file with DCB=VB,600,23200 and name it *WINFILE*. The BARR2MVS utility uses *DD:RSCWIN* as the DD name for the original Metacode print spool file and uses *DD:RSCMVS* as the DD name for the newly-created Metacode print spool file with separate records.

BARR2VB Use this utility to convert a Metacode print stream from a BARR record format to a Variable Block (VB) format.

Variable block format Metacode records are used in Docuview LFS, which lets you view a Metacode print stream, and in JES Commander, which lets you upload Metacode print streams to MVS or OS/390.

Program names

Windows BARR2VB.EXE

Syntax BARR2VB /I /O /D /P

Parameter	Description
/I	Enter the name of the input file in BARR format.
/O	Enter the name of the output file in VB format.
/D	(Optional) Include this parameter to add Docusave comments.
/P	(Optional) Use this option to specify the PrtType control group you want the utility to use from the INI file. For example, you could enter XER.

Use the /D parameter to add a dummy Docusave record to the variable block format print stream.

Your FSISYS.INI file should include a printer control group, such as PrtType:XER, which contains the options used to produce the BARR formatted Metacode print stream. This information is necessary for BARR2VB to read the BARR formatted Metacode print stream.

BARRWRAP Use this utility to reformat MVS generated Metacode output for submission to a BARR system for Xerox printing on a local area network.

Program names

Windows	BARRWW32.EXE
MVS	See the Documaker Server Installation Guide

Syntax BARRWW32 /I

Parameter Description

/I	The Metacode file. The extension must be MET, which you can omit.
----	---

NOTE You can replace the backslash (/) with a dash (-) for any parameter.

The file will be formatted for BARR spool output and the output file name will have the same name with a TXT extension.

This utility converts Metacode output from a JES2 format to a BARR format. The BARR interface attachment for Xerox Metacode printers requires Metacode print stream files to contain BARR specific information. BARRWRAP adds this information to an existing Metacode print stream file, allowing the output file to be printed via the BARR interface. After the utility is run on the Metacode file, "76 1A FF 00" will be added at the beginning of the file which informs BARR that the file is of Metacode type. A byte denoting record length is also added at the beginning and end of each record in the file.

To use this utility, you must set these INI options:

```
Environment=MVS
OutMode=JES2
```

BARRWRAP is useful when testing the GenPrint program on MVS. If the MVS system is not directly channel attached to the Xerox printer, you have to download the print data stream to a Windows system. (use no ASCII translation, but do use CR/LF). Then, using BARRWRAP, it is packaged to pass through BARR/SPOOL successfully.

NOTE Occasionally, binary data contained in the Metacode file has a sequence of hex bytes x'0D0A', which could be misinterpreted as a carriage return / line feed. This is true particularly for charts and other inline graphics. Such data streams should be BARRWRAPped on the MVS platform before being downloaded with no ASCII and no CR/LF options.

BDF2FDT Use this utility to load a complete tree, based on the specified BDF file, and then convert it to a pre-version 11.x format FORM.DAT file.

NOTE This utility is typically used by customers who have purchased tools from 3rd party vendors that parse FORM.DAT files. These tools extract a list of the forms in the library as a part of the data exchange mapping process.

Program names

Windows BDF2FDT.EXE

Syntax BDF2FDT /I /INI /O /D

Parameter Description

/I	(Optional) Enter the name of the BDF file you want to convert. If you omit this parameter, the system looks for the one in the INI file you specify using the INI parameter.
/INI	(Optional) Enter the name of the INI file which contains the name of the BDF file you want to convert. The default is FSIUSER.INI.
/O	(Optional) Enter the name of the output FORM.DAT file. The default is FORM.DAT.
/D	(Optional) Enter the effective date in YYYYMMDD format. The default is the current date.

NOTE Not all document features and options supported by Documaker Studio can be represented in a FORM.DAT file. You must make sure the forms you develop in Studio convert appropriately.

CARINTEG Use the CARINTEG utility to check the integrity of CAR archive files. This utility is built into the base system, so you may not need to run it separately if you are using version 7.5 or later.

Program names

Windows CARIGW32.EXE

Syntax

CARIGW32 /I

Parameter Description

Parameter	Description
/I	Enter the name of the CAR file.

A CAR file contains compressed NA and POL information along with a table of offsets to this information. CARINTEG queries the table of offsets to determine the number entries, counts back to the first entry in the table, then verifies that the record referenced by that offset is a valid CAR record.

CARINTEG returns a message which tells you if the CAR file is Ok or if it has errors.

CARREN Use the CARREN utility to rename CAR files. Typically, this is done to keep CAR file sizes at a manageable level.

Program names

Windows CARRNW32.EXE

Syntax

CARRNW32 /I /P

Parameter Description

/I	Enter the name of the CAR file.
/P	Enter the path in which to place the renamed file.

You must have the following two files in the working directory, along with the CAR file:

- CATALOG.DBF
- CATALOG.MDX

This utility renames the specified CAR archive file with a new, unique name. The next time archive is run, the system creates a new CAR archive file with the same name as the original CAR file.

NOTE You can automate this process using INI options.

CFA2FAP Use the CFA2FAP utility to convert a CFA or CFX file into a FAP or FXR file. CFA files are compiled FAP files. CFX files are compiled FXR (font cross-reference files). This utility is used for debugging purposes.

When you use this utility, the result matches your original FAP file, with these exceptions:

- Version records will not be decompiled
- Some error and backward compatibility issues may have been corrected when you compiled the file so the new, decompiled file will include those changes

NOTE The version of the system you use to compile the FAP and FXR files must be the same version you will use when running Documaker Server. Furthermore, the platforms must also match. For instance, if you compile the FAP and FXR files on version 11.0 for Windows, to use them in Documaker Server (GenTrn, GenData, GenPrint), you must run version 11.0 for Windows of Documaker Server.

Program names

Windows CFA2FAPW.EXE

Syntax

CFA2FAPW /I /O /F

Parameter Description

/I	The input CFA file or CFX file
/O	(Optional) The output FAP or FXR file
/F	Include this option to create a font cross-reference (FXR) file which contains only fonts

Make sure the CompiledFAP option is set ~~No~~ Yes (the default is No) in the RunMode control group of your INI files before you run the system using precompiled FAP files.

In addition, use the File, Library Setup option (or edit your INI files) to specify the path for the CompLib, which is where the system will look for the compiled files. In your INI files, you will find this setting in the following control group:

```
<MasterResource>
  CompLib=(directory or library the CFA and CFX files are stored in)
```

For MVS systems, you can specify a DD name, such as DD:COMPLIB()

NOTE If you are going to use pre-compiled FAP files, you must also use precompiled FXR files. Also keep in mind that you cannot upload CFA files.

CPCNV Use the CPCNV utility to convert the text of a flat file, record by record, from a source codepage to a destination codepage. This utility supports record lengths up to 32k.

NOTE You can port this utility to other platforms.

Program names

Windows	CPCNVW32.EXE
MVS	See the Documaker Server Installation Guide

Syntax CPCNVW32 /S /D /I /O /N /R

Parameter Description

/S	The codepage the source file is currently written in.
/D	The code page to convert to.
/I	The text file to convert (you can use asterisks as wildcards).
/O	(Optional) The name of the output file.
/N	(Optional) A nontranslatable character value, the default is 255.
/R	(Optional) Reads the CR/LFs in the ASCII file and converts the data into multiple records. Use this parameter if you use FTP or similar communications programs to upload files. If the program you use to upload the files converts CR/LFs into separate records, you can omit this parameter.

Use the CPCNV utility to convert text files written in one codepage to another codepage. The CPCNV utility loads either the FSISYS.INI or FAPCOMP.INI file to find these FMRES control group options:

```
< FMRes >
  DefLib = ..\MSTRRES\FMRES\DEFLIB\ (default shown)
  Codepage = CODEPAGE.INI (default shown)
```

This utility uses the CODEPAGE.INI file in your \mstrres\fmres\deflib directory. If the file is not there, the utility displays an error message. You can specify a different location by adding an INI option in the FSISYS.INI or FSIUSER.INI files, as shown here:

```
< FMRes >
  DefLib = the path where the codepage.ini is located.
```

The CODEPAGE.INI file contains information about characters in various codepages. Here is an excerpt of the CODEPAGE.INI file:

```
< Codepages >
  Codepage    = 1004,W1
  Codepage    = 863,CF
  Codepage    = 850,PM
  Codepage    = 437,PC
  Codepage    = 37,Z1
< Codepage >
  Char        = SP010000,          space, , 32, 32, 32, 32, 64
  Char        = SP020000,          exclam, , 33, 33, 33, 33, 90
  Char        = SP040000,          quotedbl, , 34, 34, 34, 34,127
  (and so on)
  Char        = LA160000,          Acircumflex, ,194,132,182, 0, 98
  Char        = LA150000,          acircumflex, ,226,131,131,131, 66
  (and so on)
```

The first control group, CODEPAGES, lists the codepages specified by this file. The two character abbreviation, following the codepage number, specifies an internal character set name used by some system font conversion utilities.

The second control group, CODEPAGE, specifies character names and their associated code points in the different codepages listed in the first control group. The first column is the name of the character when printing to an AFP printer. The second column is the name of the character when printing to a PostScript printer. The third column is the name of the character when printing to a TrueType printer (not currently supported). The remaining columns correspond to the code points for the codepages specified in the first control group.

In the example above, there are five codepages specified: 1004, 863, 850, 437, and 37. Therefore, the last five columns represent code points for codepages 1004, 863, 850, 437, and 37 respectively.

For example, the first character defined (the space character) has these attributes:

```
Char        = SP010000,          space, , 32, 32, 32, 32, 64
AFP name    = SP010000
PostScript name = space
TrueType name = (blank, not used)
1004 code point = 32
863 code point = 32
850 code point = 32
437 code point = 32
37 code point= 64
```

where as the A-circumflex (Â) character has the following attributes:

```
Char        = LA160000,          Acircumflex, ,194,132,182, 0, 98
AFP name    = LA160000
PostScript name = Acircumflex
TrueType name = (blank, not used)
1004 code point = 194
863 code point = 132
850 code point = 182
437 code point = 0 (not defined for this codepage)
37 code point = 98
```

Therefore, if you were to convert a text file built using codepage 863 (Canadian French) to codepage 1004 (ANSI/Skywire Software standard), any space characters would remain unchanged but any A-circumflex characters would change from a 132 to a 194 code point.

See Using Fonts, in the [Documaker Server System Reference](#) for more information about fonts and codepages.

Understanding the System

Sometimes you need to transfer files between the PC and host (MVS) platforms. There are a number of products that provide this capability. For instance, you can use FTP to transfer files from the PC to the host (MVS). FTP can transfer a file using *binary* or *text* mode.

- Binary mode means do not translate the characters contained in the file.
- Text mode means translate the characters from ASCII on the PC to EBCDIC on the host.

Text mode will also write a separate record on the host for each carriage return line feed combination found in the original PC file.

Unfortunately, the Text mode ASCII to EBCDIC translation used by FTP does not match our standard ASCII to EBCDIC translation for extended ASCII characters (code points 128 and above contain international characters, some currency and punctuation symbols, and so on). Therefore, if system resource files to be uploaded contain some of these extended ASCII characters, the files must be uploaded as Binary.

If a system resource file, such as a FAP, FXR, or DDT file is uploaded as binary, not only do you need to convert the ASCII characters to EBCDIC, you also need to create records that correspond to each line of text in the PC file.

On the PC, the end of a line is indicated by a couple of control characters, specifically a carriage return character followed by a line feed character (CRLF). Since the PCL file was uploaded as binary, the file still contains the CRLF characters. Use the /R parameter to treat the CRLFs as an end of line indicator when converting the file on MVS.

Some products, including IBM's Personal Communication, provide the ability to upload a file as binary but will translate CRLFs into separate records when uploading the file. Do not use the /R parameter if you use a file transfer program that converts lines ending with CRLFs into separate records when uploading to MVS.

CSET2FAP Use the CSET2FAP utility to convert Document Sciences CompuSet scripts into Skywire Software FAP files. This utility does conversions as part of a batch process.

NOTE Using this utility reduces the work necessary to convert a Document Sciences resource base into a Skywire Software resource base. These commands are supported:

Command Description		Command Description	
BD	bold	BL	blank line
BOX	create a box	CB	change baseline
CC	center on column	COLOR	set color
CP	center on page	CW	column width
DL	dot leader	EL	end underline
F	select font	HR	horizontal rule
JU	horizontal justify	LT	non bold
NL	new line	NP	new page
NPR	new page recto	PA	set paragraph indent
PD	page depth (height)	PT	point size
PW	page width	QL	quad left
QR	quad right	SK	skip text (comment)
SP	space vertically	T	select tab
TABC	set text-centered tab	TABJ	set text-justified tab
TABL	set quad left tab	TABR	set quad right tab
UL	underline	VR	vertical rule

Keep in mind the utility does not convert the full set of CompuSet logic.

Syntax CS2FPW32 /I /C /F /S

Parameter Description

I	The name of the INI file from which to load the INI options discussed in the configuration topic following this table. The default is the FSISYS.INI file.
C	The name of the configuration file that contains font information. If you omit this parameter, the utility uses the value for the Config option in the CompuSet control group.
F	The name of the form you want to convert.
L	Including this parameter tells the utility to log all of the debug information created when creating FAP files from a CompuSet file. This parameter is turned off by default.

Parameter Description

S	The name of the file that contains the master styles, if any. If you omit this parameter, the utility uses the value for the MasterStyles option in the CompuSet control group.
---	---

All parameters are case insensitive and can be preceded by a slash (/) or a dash (-).

Configuring the INI File

Since Skywire Software applications can use the same font files as the Document Sciences system, first copy the font files into your Skywire Software Metacode resource directory. This is the same place the Metacode fonts licensed by Skywire Software are located. Then use Font Manager to import the fonts. Specify the name of the new font cross-reference (FXR) file in the XRFFile option of the MasterResource control group. The CSET2FAP utility also uses these INI options:

```
< CompuSet >
  MasterStyles =
  Config       =
  ExtremeLogging =
< MasterResource >
  DefLib       =
< Config:XXX >
  XRFFile      =
< Loaders >
  Loader       =
< Loader:CompuSet >
  Desc         =
  Func         =
  Module       =
```

Option Comments

CompuSet control group

MasterStyles	(Optional) Enter the global styles definition file name.
Config	Enter the name of the file that contains logical and machine font definitions and the font files in which they are located.
ExtremeLogging	Enter Yes to view all of the debug information when creating a FAP file from a Document Sciences file. The system defaults to No.

MasterResource control group

Deflib	Enter the path to your Skywire Software resources.
--------	--

Config:XXX control group where XXX is the name of the library or workspace

XRFFile	Enter the name of the font cross-reference file (FXR) you want the system to load.
LOGO.DAT	Enter the name of the file containing the fonts you want to use as logos. For more information see Using Fonts as Logos on page 59

Option	Comments
--------	----------

Loaders control group

Loader	This entry is used to locate image loader INI groups. For instance, you could enter <i>CompuSet</i>
--------	---

Loader:CompuSet control group

Desc	Enter the description you want to appear in the file types list of the File Open window. For instance, you could enter <i>CompuSet files (*.TXT)</i>
Func	Enter the name of the image loading function, such as <i>CompuSetImageLoader</i>
Module	Enter the name of the image loader module, such as <i>SLDRW32</i>

NOTE Because the conversion does not handle all CompuSet commands, you may receive some error messages.

Using Fonts as Logos

If you have fonts you use as logos, you must first convert them to LOG files. Then, enter the root file name for each logo in the LOGO.DAT file. The LOGO.DAT file should reside in the current working directory. The LOGO.DAT file is a semicolon-delimited text file that names the various rotations of logo fonts and it should look similar to this:

```
[file name for 0° rotation];[file name for 90° rotation];[file name for 180°rotation];[file name for 270° rotation];
```

Any of the rotations can be left out, but you must include the semicolon as a delimiter.

NOTE The logo fonts should not be present in the FXR since the utility uses not finding a font as a trigger to look for a logo.

CVTFASR Use the CVTFASR utility to convert a FORMDEF and SETRCPTB pair into BDF (Business unit definition file), GRP (group file), and FOR (form file) files. The existing FORMDEF and SETRCPTB files are not modified or removed.

Syntax: CVTFASR /FORMDEF /SETRECIP /INI

Parameter Description

FORMDEF Enter the name of the form definition file.

SETRECIP Enter the name of the set recipient file.

INI Enter the name of the INI file.

The CVTFASR utility reads the specified FORMDEF and SETRECIP files and generates the appropriate BDF, GRP, and FOR files. If you include the INI parameter, the utility opens and reads that INI file. The utility tries to locate these INI options to get the directory locations into which the BDF, GRP and FOR files will be written.

```
< MasterResource >
  BDFLIB = .\bdfplib
  GRPLIB = .\grplib
  FORLIB = .\forlib
```

If the INI option for one or more of the generated file types is missing, the utility writes the files for that type into the working directory.

Here is an example of the messages you may see when running the CVTFASR utility on Windows:

```
c:\fap\mstrres\rpex1>cvtfasrw /formdef=.\deflib\form.dat /
setrecip=.\deflib\setrcptb.dat
--- DocuCorp CVTFASR Utility Program (C) ---
--- Convert Formdef And SetRecip Files ---

Number Of Files Generated From Formdef & SetRecip:
Basedef (BDF) : 1
Group (GRP) : 4
Form (FOR) : 24

CVTFASRW was successful.
```

DALRUN Use this utility to execute and debug DAL functions.

Program names

Windows DALRW32.EXE

Syntax

DALRW32 /X /INI /D /T

Parameter Description

/X	(Optional) Supplies the name of a script to run. If you omit this option, you can use this INI option to provide the name of the script: <pre>< DALRun > Script = file name</pre> You can use any extension. The default is <code>.DAL</code>.
/INI	(Optional) Supplies the name of an INI file to load. This INI file supplies additional parameters and options. If an INI file named DALRUN.INI is present, the utility loads it by default. Here are the INI options you can include in the INI file: <pre>< DALRun > Title = title string(an override to the window title) Script = file name (the script to run) < DALFunctions > Keyword = DLLMOD->FunctionName Keyword2 = DLLMOD->FunctionName2 (and so on)</pre>
/D	The debug switch starts the DAL debugger. When on, the script executes in single step mode and registers this DAL function <code>DEBUG("message")</code>. The <code>DEBUG</code> function breaks execution, displays a message, and invokes the debugger in single step mode.
/T	This parameter sends certain text messages to the standard output device. These messages are not visible at runtime, but may be redirected when you run this utility. Here is an example: <pre>DALRW32 /ini=test /d /t > test.txt</pre>

NOTE Refer to the [DAL Reference](#) for information on the DAL functions you can use to create the scripts DALRUN executes.

Debug messages, certain errors, and a dump of the symbol table at the end of the run are examples of output this utility will generate.

DB2DB Use this utility to copy data from one database table to another database table. The database tables can be from a different DBMS, such as one from ODBC and the other from a native DB2 database.

Program names

Windows DB2DBW32.EXE

Syntax

DB2DBW32 /ST /SD /SD /TT /TD /TH /?

Parameter	Description
ST	Enter the name of the source table.
SD	Enter the name of the DFD file for the source table.
SH	Enter the type of handler for the source table.
TT	Enter the name of the target table.
TD	Enter the name of the DFD file for the target table.
TH	Enter the type of handler for the target table.
?	Print the parameter usage information.

NOTE The source and the target table names must be different. You can rename target table after the conversion to a new database. For example:

```
db2dbw32 /st=xdb ..... /tt= xdb ...    (incorrect)
db2dbw32 /st=xdb ..... /tt= xdb2 ...    (correct)
```

Here are two examples:

Example 1 How to convert an XDB dictionary (a CB5 table) into a DB2 table:

```
db2dbw32 /st=xdb /sb=.\deflib\xdb.dfd /sh=CB5 /tt= xdb2 /td=
.\deflib\xdb.dfd /th=db2
```

These parameters tell the utility to copy the source table (XDB), which is associated with the xBase DBHandler called *CB5* and is mapped using the XDB.DFD file in the .\deflib directory, to the target table (xdb2), which is associated with a DB2 DBHandler called *DB2* and is also mapped using the same DFD file. In this example, the actual name of the XDB2 DB2 table is *CP_Xpress*.

In this example, your FSIUSER.INI file contains entries similar to these. The DB2 table, cpxdb2, is contained in the DB2 database, CP_Xpress.

```
< DBHandler:DB2 >
Database = CP_Xpress
BindFile = c:\dap32103\dll\db2lib.bnd
Connect   = Yes
CreateTable = Yes
UserID    = db2admin
Debug     = Yes
```

```

        PassWd      = XXX
< DBTable:xdb >
        DBHandler = CB5
< DBTable:xdb2 >
        DBHandler = DB2
< DB2_FileConvert >
        xdb2       = cp_xpress_xdb

```

You can use the DB2_FileConvert INI control group to specify longer names for the table in DB2.

Option	Description
CreateTable	Enter Yes so DB2 will create the table if it does not exist.
BindFile	Enter the name and path of the DB2LIB.BND file.

Example 2 How to convert an Access table into a DB2 table:

```

db2dbw32 /st=coverage /sd=.\deflib\coverage.dfd /sh=CB5 /tt=
coverage2 /td= .\deflib\coverage2.dfd /th=db2

```

These parameters tell the utility to copy the Microsoft Access source table (coverage), which is associated with a DBHandler called **ODBC** and is mapped using the COVERAGE.DFD file in the .deflib directory, to the target table (coverage2), which is associated with a DBHandler called **DB2** and is mapped using the COVERAGE2.DFD file in the .deflib directory.

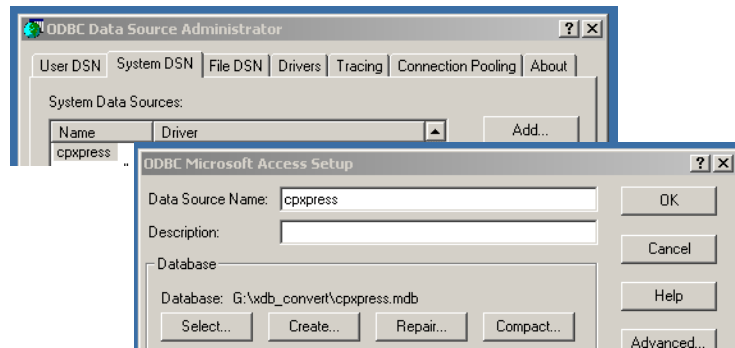
In this example, your FSIUSER.INI file contains entries similar to these. The DB2 table, coverage2, is contained in the DB2 database, CP_Xpress.

```

< DBHandler:DB2 >
    Database = CP_Xpress
    BindFile = c:\dap32103\dll\db2lib.bnd
    Connect   = Yes
    CreateTable = Yes
    UserID    = db2admin
    Debug     = Yes
    PassWd    = MVF
< DBHandler:ODBC >
    Server    = cpxpress
< DBTable:Coverage >
    DBHandler = ODBC
< DBTable:Coverage2 >
    DBHandler = DB2

```

These screen samples are an example of the MS Data Sources (ODBC) setup from a conversion:



DCD2FAP Use this utility to convert a Docucorp Compound Document (DCD) file into a FAP file. DCD files are produced by Skywire Software applications such as Documerge for Windows NT version 3.0 and older versions of Documanager (Printcommander for Windows 95 version 2.0.)

Program names

Windows DCD2FAPW.EXE

Syntax DCD2FAPW /I /X

Parameter Description

/I	The name of the input DCD file. You can use wildcards.
/X	The name of the font cross-reference (FXR) file. Include the extension (.FXR).

The resulting FAP file will have the same name as the DCD file but with a .FAP extension.

DFD2DDL Use this utility to generate DDL (Data Definition Language) files from your DFD files. DDL files consist of the SQL statements that create your tables and views. This is helpful if you have an RDBMS database where an administrator is required to create tables and indexes and users are not typically granted those privileges.

Program names

Windows DFD2DDLW.EXE

Syntax

DFD2DDLW /I /O /D /T /S /P /L

Parameter Description

/I	Enter the name of the DFD file. Include the full or relative path.
/O	Enter the name you want the utility to assign to the DDL file it will create. The default is the DDLFile root plus the extension SQL .
/D	Enter the name of the target database. You can choose from: MSSQL, DB2, Oracle, MySQL. The default is MSSQL.
/T	Enter the name of the target table. The default is the DFD root name.
/S	(Optional) Enter the name of the target schema. This is typically used for Oracle schema, DB2 schema, and MSSQL dbowner.
/P	(Optional) Enter the name of the primary table index. This allows for unique constraints on key fields.
/L	(Optional) Enter the name of the target location. This is used for DB2 (location) and MySQL (database name).

You should avoid the following ExtTypes because they are not supported as external data types in Documaker's database handlers:

- CHAR
- UCHAR
- DECIMAL
- DOUBLE
- LONG_DOUBLE
- TIMESTAMP
- DATETIME

FAP2CFA Use the FAP2CFA utility to convert a FAP file into a compiled FAP file. By pre-compiling your FAP files, you can speed processing.

NOTE The version of the system you use to compile the FAP and FXR files must be the same version you will use when running Documaker Server. Furthermore, the platforms must also match. For instance, if you compile the FAP and FXR files on version 11.0 for Windows, to use them in Documaker Server (GenTrn, GenData, GenPrint), you must run version 11.0 for Windows of Documaker Server.

Program names

Windows FAP2CFAW.EXE

Syntax

FAP2CFAW /I /O /INI /F

Parameter Description

/I	The input FAP file or font cross-reference (FXR) file (you can use asterisks as wildcards).
/O	The output CFA or CFX file. You can enter an absolute (d:\dap\myinc\mstres\fab) or relative path (..\mstres\fab).
/INI	Specifies an INI file which contains additional parameters. You can enter an absolute (d:\dap\myinc\mstres\fab) or relative path (..\mstres\fab).
/F	Include this option to create a compiled font cross-reference (FXR) file which contains only fonts.

This utility lets you compile FAP files one at a time. To compile all of the FAP files listed in a FORM.DAT file, see [FAP2CFA on page 83](#). To convert a CFA or CFX file back into a FAP or FXR file, see [CFA2FAP on page 53](#).

Make sure the CompiledFAP option is set **Yes** (the default) in the RunMode control group of your INI files before you run the system using precompiled FAP files. In addition, use the File, Library Setup option (or edit your INI files) to specify the path for CompLib, which is where the system will look for the compiled files. In your INI files, you will find this setting in the following control group:

```
< MasterResource >
  CompLib = (library which contains the CFA and CFX files)
```

For MVS systems, you can specify a DD name, such as DD:COMPLIB()

NOTE If you are using pre-compiled FAP files, you must also use compiled FXR files.

Example

If you enter this command...	The result is a compiled FAP file named...
FAP2CFAW /l=qaihead	qaihead.cfa
FAP2CFAW /l=qaihead / o=qaihead1	qaihead1.cfa
FAP2CFAW /l=cwqa240.fxr /f	cwqa240.cfx
FAP2CFAW /l=cwqa240 /f	cwqa240.cfx
FAP2CFAW /l=qaihead.old	qaihead.cfa
FAP2CFAW /l=..\faps\qaihead	qaihead.cfa, located in the ..\fap subdirectory

FAP2DDT Use the FAP2DDT utility to create or update a DDT file from a FAP file.

NOTE You can also perform this task using the File, Convert, Multiple FAPs option in Docucreate.

Program names

Windows FAP2DDTW.EXE

Syntax

FAP2DDTW /I /O

Parameter Description

/I	Enter the name of the input FAP file. Omit the extension.
/O	(Optional) Enter the name of the DDT file. If you omit this file name, the utility uses the FAP file name with the DDT extension. If the DDT file does not exist, the utility creates it.
/X	Enables the output of <i>X;Y;FontID;</i>
/V	Enter 7 or 8, force output to the new (8) or old (7) format, defaults to the same format as DDT input, or version 8.0 format if you omit this parameter.
/S	Synchronizes with the existing DDT file.

The FAP2DDT utility creates a data definition table (DDT) file from the existing FAP. The DDT file stores image rule assignments. The parameters are optional and case insensitive.

Keep in mind the FAP2DDT utility adds a default SetOrigin rule to the Image Rules section if one does not exist. The SetOrigin rule that is added is shown here:

```
;SetOrigin;Abs+0,Abs+0;
```

For the FAP2DDT utility to add a SetOrigin rule into your DDT file, include this INI option in the FAPCOMP.INI file:

```
< DDTResource >
AutoIncludeSetOrigin = ;SetOrigin;Abs+0,Abs+0;
```

The DDTResource control group is used for DDT settings, such as WriteCoordinates, MultipleDDTs, and so on.

A SetOrigin rule is only added to a DDT file if there is an AutoIncludeSetOrigin option and these conditions are true:

- The AutoIncludeSetOrigin setting begins with a semicolon (;)
- The AutoIncludeSetOrigin setting contains the string "SetOrigin"
- The DDT file does not already contain a SetOrigin rule (SetOrigin, SetOriginI, SetOriginM, RULSetOrigin2, UTILSetOrigin)

Other than verifying that the rule begins with a semicolon and contains some form of the string "SetOrigin," no other validation is performed. You must make sure you specify a valid rule with valid parameters.

NOTE Choosing the Convert Multiple FAPs, Update DDT from FAP option in Docucreate is another way to have the default SetOrigin rule added to the DDT file if it does not exist. The same caveats apply.

Example FAP2DDTW /l=fapfile

This will create a file named FAPFILE.DDT whose contents look like:

```
<Image Rules>
;SetImageDimensions;0,0,26400,20400,400,600,400,600;

/* By default, this image contains the following fields */
<Image Fields>

<Image Field Rules Override>
;0;0;BLANK FIELD NAME;;;BLANK FIELD NAME;;0;;noopfunc;;;;;
```

FAP2FRM Use the FAP2FRM utility to compile a FAP file into a Xerox FRM Metacode overlay file.

Program names

Windows FAP2FRMW.EXE

Syntax FAP2FRMW /I /X

Parameter Description

/I	Enter the name of the FAP file. Omit the extension.
/X	Enter the name of the font cross-reference (FXR) file. Omit the extension.
/P	Enter the INI file PrtType to use, such as AFR XER
/D	Enter SF for short bind front page duplex. Enter SB for short bind back page duplex.
/H	(Optional) Add this parameter to use HMI.
/C	(Optional) Add this parameter to use color in the FRM file.

NOTE The name of the FAP file cannot exceed six characters (123456.fap).

The FAP2FRM utility requires these files:

- FSISYS.INI
- Font cross-reference file, such as REL103.FXR

FAP2FRM generates a Xerox form printer resource file (FRM) from a FAP file. In addition to the FAP file name, the FXR font cross-reference file used by the FAP must also be specified. Since FRM file is a printer resident resource, the file name should be no more than six characters long.

FAP2FRM looks for a control group named Printer in the FSISYS.INI. In the Printer control group, FAP2FRM looks for the PrtType option, which determines the type of printer being used, such as AFP, XER, or PCL. Use the /P parameter to specify which PrtType control group the utility should use. The default control group is PrtType:XER.

If the FAP file contains multiple pages, FAP2FRM generates multiple output files with 2-digit numeric suffixes. In this case, file names for these FAP files should not exceed four characters (1234.FAP).

After running FAP2FRM, download the output FRM files to the printer, using the XERDNL utility. For more information, see [XERDNL on page 206](#)

Also, in the Form Set Manager, check the Printer Resident field for each such image.

If the FAP file contains logos, the FAP2FRM utility produces a Xerox FRM file using either Xerox fonts or Xerox images, based on the value for the ImageOpt option in your Xerox printer control group. If you set the ImageOpt option to Yes, Xerox images (IMG files) are used in the Xerox FRM file the utility produces. Otherwise, Xerox fonts (.FNT files) are used.

NOTE Image references are supported in FRM files. If you set the ImageOpt entry to Yes, the system generates the required GHO headers and packets in the FRM file so FAP files that contain IMG references can be converted into Xerox FRMs.

Example Follow these steps to convert a precompiled MET file into a printer-resident FRM file that will be used in the form set. These instructions assume you are converting an existing MRL that used multiple floating page segments (METs) on some pages:

- 1 For performance reasons, when faced with a choice of multiple floating page segments on a page, choose the largest one (the one with the most data on it (characters, lines, or shading) to be the FRM so that image can be resident on the printer and does not have to be in the printed output.
- 2 Change all references to that image in the FORM.DAT file, making that image printer-resident. You can do this using the Form Set Manager.
- 3 In the Image Editor, edit the FAP file for that image. If it was used as a floating image (precompiled MET), then it is probably not positioned correctly on the page to be a FRM, because FRMs are static and should be thought of as whole-page images, not floaters. Position the image to where it should be on the page, then use the Format, Page Properties option to select **Letter** as the paper type.
- 4 Next, select Format, Image Properties, and click the Load DDT button. On the Image Rules tab, highlight **SetOrigin**, and look at the Data field. Because the image was probably a floater, it should have relative positioning, such as **REL+0,REL+0**. Change it an absolute value, such as **ABS+0,ABS+0**. Save the image and DDT information. You do not have to change any FAP or DDT files for images that are positioned before where the FRM image is positioned on the page.
- 5 Next, load the next image that appears directly after the FRM image on that page in the Image Editor. Change its DDT information similarly, except position it below the FRM image, with coordinates larger than zero.

You can get good coordinates from the standard NAFILE that was used before making these changes. If there are any images that occur after this one on the same page, you should not have to change them, because they should be already set to relative positioning. If the system did not force the image immediately after the FRM to be absolutely positioned on the same page, triggering would tell the GenData program to insert a page break at the end of the FRM, and the next image will incorrectly be printed at the top of the next page. After you do this, you can run the GenData program and make sure the images are positioned correctly.

- 6 Create the FRM using the FAP2FRM utility, then download the FRM to the printer. You can sample it on the printer to make sure it prints and has the correct positioning on the page. Keep all FRM files on the workstation also in FormLib when the GenPrint program runs.
- 7 Finally, run the GenPrint program and print.

Limitations You cannot have FRM and MET files in the same MRL which have the same file name. For instance, in RPEX1's FORM.DAT, page one has image Q1SNAM listed as printer-resident (an FRM). Page two has the same file name Q1SNAM, only for this page it is a pre-compiled MET. Doing so will cause errors.

FAP2MET

Use this utility to compile a FAP file into a pre-compiled Xerox Metacode overlay. This utility generates a Metacode print file from a FAP file.

Program names

Windows	FAP2METW.EXE
MVS	See the Documaker Server Installation Guide

Syntax

FAP2METW /I /X /P /N /VF /SV /D /H /C /LB

Parameter Description

/I	Enter then name of the FAP file (you can use asterisks as wildcards).
/X	Enter the name of the font cross-reference (FXR) file.
/P	Enter the INI file PrtType to use, such as REL REL
/N	Enter the font ID to use to print name of image, such as 11006.
/VF	Add this parameter to sample print with variable fields.
/SV	Add this parameter to save the MET file for the GenPrint program. Keep in mind you cannot use this parameter if you have the ImageOpt INI option set to Yes for floating images which contain inline graphic logos.
/D	(Optional) Enter SF for short bind front page duplex, ente SB for short bind back page duplex.
/H	Add this parameter to use HMI.
/C	Add this parameter to use color in the MET file.
/LB	Add this parameter to override the default logical bottom for the form being normalized. To specify a different logical bottom value, include the parameter and a value, as shown here: /LB = value

This utility requires these files:

- FSISYS.INI
- Font cross-reference file such as REL103.FXR

Depending on the option flags, the print file may be a print-ready file or a pre-compiled file for use with the GenPrint program.

In addition to the FAP file name, you must also specify the name of the font cross-reference file (FXR) used by the FAP file. To print the file, you must install Xerox versions of the fonts used in the FAP file on the printer.

NOTE The INI options for FAP2MET are in XEROXJDL control group in the FSISYS.INI file in version 8.0, and in PrtType:??? control group in versions 8.5 and later.

This table shows the INI options which will work in the FSI environment:

Option	Setting
DJDEIden	A'@@@DJDE'
DJDEOffset	0
DJDESkip	8
JDLHost	IBMONL
JDLData	0,255
JDLCode	NONE
JDLName	DFAULT
JDEName	DFLT
JDLRStack	0,10,X'13131313131313131313'
JDLROffset	0,9,X'121212121212121212'
JDLRPage	1,5,X'FFFF26FFFF'
ImageOpt	No
PrinterInk	Blue
Environment	Win
OutMode	BARR

NOTE These settings must match those set in the JDL. The JDL is a compiled JSL installed on the Xerox printer as a resource. The JDL contains the print instructions for Xerox print jobs.

Running on MVS If you are running the utility on MVS, set the Environment and OutMode options to:

```
Environment= MVS
OutMode= JES2
```

This is based on definitions in the FSI SYS.INI file. For example, in the INI file, you can set up different modes of Xerox printing by setting up different control groups, such as...

```
< PrtType:XER1 >
  DJDEIDEN = E'$$Xerox'
  OUTMODE = JES2
.....
< PrtType:XER2 >
  DJDEIDEN = A'@@@DJDE'
  OUTMODE = BARR
.....
```

When you want to send the FAP2MET output to the printer via JES2, you could provide /P= XER1, or if your output is going through BARR, you could provide /P = XER2 to select proper paragraph. If you omit the /P parameter, it defaults to XER.

The `/D` parameter provides short edge binding duplex function. `D=SB` means the image is to be compiled for short edge binding as a back page. This forces the text to be printed in inverse portrait mode. Your FXR file should have the corresponding font names suitable for inverse portrait mode. `D=SF` means the image is to be compiled for short edge binding as a front page. The default is `SF`.

The `/V` parameter creates Metacode which can be sent to the printer immediately for sample printing, with all field positions indicated with XXXX strings. The name of the form is printed at the bottom right corner, using the font ID you specified with the `/F` parameter. Check your FXR file, and make sure the font ID is assigned to a suitable font.

NOTE Do not use this parameter if you are precompiling Metacode files in preparation to run the GenPrint program.

The `/SV` parameter creates an intermediate form of Metacode which can not be sent to the printer directly, but which at GenPrint time, gets merged with field data and gets converted into proper Metacode. This combined Metacode is then sent to the printer. Therefore, if you are creating the Metacode files as precompiled Metacode files to be used in GenPrint process, use `/SV` parameter. In the absence of either of these flags, a sample print Metacode is generated without the XXXX strings.

If you add `/H` to either the `/V` or `/SV` parameters (`/VFH` or `/SVH`), the system generates a more efficient Metacode output by combining several Metacode records into one, reducing the output file size. You can also specify this option as a separate standalone parameter (`/H`).

NOTE This feature is available only for portrait mode printing.

Use the `/C` parameter when you want to print the compiled instream Metacode on a 4850 or 4890 highlight color printer, and some elements of the form have a color specification. (Be sure to set PrinterInk option to whatever ink is installed on the printer. Also, the SendColor option must be set to `yes`).

NOTE The PrinterInk option supports red, blue, green, ruby, violet, brown, gray, cardinal, royal, cyan, and magenta.

Parameters passed via
the PARM=' 'field (MVS/
JCL)

In earlier versions (before 8.5), these parameters were passed in the PARM=' 'field: `fapfilename`, `SAVE` or `TEMPLATE` and `USEHMI`.

The parameters are identical in both environments. The only difference is that an extra backslash (`/`) must precede the list of parameters. A typical EXEC statement in a FAP2MET run JCL would be:

```
//FAP2MET      EXEC      PGM=FAP2MET,PARM=' /I=fapfile /P=XER /VFH /C'
```

Producing normalized Metacode

You can use the FAP2MET utility to produce normalized Metacode files. To produce a normalized Metacode file, you must add these INI settings to your FISISYS.INI file:

```
< Control >
    Normalize = Norm
< PrtType:Norm >
    ...
```

Option	Description
Normaliz	Specify the PrtType control group you want the system to use when it normalizes the files. For instance, if you create a control group called PrtType:Norm to contain the option you want the system to use when it normalizes Metacode files, enter <i>Norm</i> . If the control group you specify is missing or does not appear to be a Xerox printer group (Class=XER) you will receive errors.

NOTE Typically, the Class option is not explicitly set in the INI file. The Class name is normally derived from the first three letters of the Module option. If the Module is not XERW32, and you want the system to think the control group is a Xerox print group, you must set the Class option to XER.

You can also use the /LB parameter to override the default logical bottom for the form being normalized. To specify a different logical bottom value, include the parameter and a value, as shown here:

```
/LB = value
```

NOTE You only use the /LB parameter when normalizing files.

GenPrint program notes

The INI options required for the GenPrint program are the same as those needed by the FAP2MET utility. In addition, there are other parameters which affect printing on Xerox printers. In the Print control group, or in the PrtType:XER control group,

```
CompileInstream = Yes or No
```

and in the RunMode control group,

```
DownloadFAP = Yes or No
```

The CompileInStream=Yes and DownloadFAP=Yes options load all of the FAP files in memory when the GenPrint program runs. This is slow, and is typically used only during testing and development. This mode does not need precompilation of FAP files into intermediate Metacode files.

For production environments, use the CompileInStream=No and DownloadFAP=No options. This mode requires that you precompile the FAP files using the FAP2MET utility with the /SV parameter.

The precompiled Metacode files must be available to the GenPrint program in the MSTRRES\FORMS directory in Windows, or in a PDS which is assigned to the PMETLIB DD statement on MVS.

Multi-page FAP files The FAP2MET utility supports multiple page FAPs. For example, if TEST.FAP contains 10 pages, FAP2MET will build 10 pre-compiled met files. These files will be used by the GenPrint program as each page is printed.

When you run the FAP2MET utility on a multi-page FAP file, the utility creates a FAP file for each page. In the DDT file, you should then make an entry for the FAP file, and add an eject page for each additional page that makes up the form. The system knows by the eject pages to look for additional FAPs for this form.

You can also use the FAP2MET utility to produce a test print version of a FAP file, otherwise known as a print-ready file. This file is only to be sent to the printer. It is not to be used with the GenPrint program. This type of file is produced when you omit the /SV parameter.

FAP2OVL Use the FAP2OVL utility to compile a FAP file into an overlay for an AFP printer.

NOTE You can also create overlays using the Image Editor's Tools, Compile option. See the [Docucreate User Guide](#) for more information. You can also use the OVLCOMP utility. See [OVLCOMP on page 174](#) for more information.

Program names

Windows	FAP2OVLW.EXE
MVS	See the Documaker Server Installation Guide

Syntax FAP2OVLW /I /X /VF /O /C /R /INI /NO

Parameter Description

/I	The FAP file you want to compile. Omit the extension. You can use asterisks as wildcards to select multiple files.
/X	The font cross-reference (FXR) file. Omit the extension. You can enter an absolute (d:\dap\myinc\mstres\rel110) or relative path (..\mstres\rel110).
/VF	Add this parameter to print variable fields.
/O	Specifies the output directory. You can enter an absolute d:\dap\my\mst\ovllib) or relative path (..\mstres\ovl).
/C	Add this parameter if the AFP overlay should use color. Enter one of the named AFP colors such as blue, red, magenta, green, cyan, yellow, dark_blue, orange, purple, dark_green, dark_cyan, mustard, gray, or brown.
/R	Enter 240, 300, or 600 to indicate the resolution in dots per inch (DPI). The default is 240. Use this parameter to support producing AFP overlays using 240, 300, or 600 DPI for coordinates and bitmap data produced when using inline logos. Image Editor and Documaker Studio can produce AFP output (print streams and normalized AFP files) using either a 240 or 300 DPI coordinate system.
/INI	Specifies an INI file which contains additional parameters. You can enter an absolute (d:\dap\myinc\mstres\fsisys.ini) or relative path (..\mstres\fsisys). The file name does not have to be FS/SYS.INI .
/NO	Add this parameter if you want the FAP file name at the bottom but no fields.

NOTE This utility is not case sensitive. You can use dashes (-) instead of slashes (/) to separate parameters.

The FAP2OVL utility generates an AFP OVL overlay printer resource file from a FAP file. In addition to the FAP file name, you must also specify the font cross-reference (FXR) file used by the FAP file.

The /VF parameter prints the FAP file name at the bottom of the page using font ID 3001, and fills all variable fields on the FAP with characters. The /NO option only prints the FAP file name at the bottom of the page using font ID 3001. These options are useful when you are developing and proofing your implementation.

NOTE Do not use the /VF or /NO options when you are creating an overlay (OVL) file for production print use.

After a production quality overlay file is generated, you have to make it available to the PSF2 by copying it to the print server and add it to the PSF2/ Librarian database.

On MVS, the overlay has to be copied into a PDS, which is attached to the OVERLIB DD statement in the PSF PROC, or a PDS which is assigned to the USERLIB= parameter, on the OUTPUT statement in the job JCL. After all the required overlays are installed on PSF2 or PSF, run the GenPrint program with this INI option:

```
SendOverlays = Yes
```

The spool file the GenPrint program creates will include calls to merge those overlays as needed.

Printing in color

Include the /C parameter if the AFP overlay should print in color. AFP highlight color printing on printers from Xerox and Océ is supported. Before using this feature, make sure the:

- SendColor INI option is set to Yes.
- Objects you want to print in color (text, lines, shades, and so on) are set to print in color. The Print in Color option is on the Color Selection window in the Image Editor. You can display this window by clicking the Color button on the object's Properties window.

The system maps the RGB (red,green,blue) color setting for each object to the closest AFP named color. The default AFP named colors are blue, red, magenta, green, cyan, yellow, dark_blue, orange, purple, dark_green, dark_cyan, mustard, gray, and brown.

Use the NamedColors option to tell the system to use only specific AFP named colors:

```
< PRTType:AFP >
NamedColors = blue
```

For example, if you wanted all highlight (non-black) colors mapped to blue, you would set the NamedColors option to blue, as shown above.

To allow the mapping of the colors you assigned to the objects in the FAP file to multiple colors, separate each color with a semicolon (;). For example, to use all of the default AFP named colors except brown, set the NamedColors option as shown here:

```
NamedColors = red;blue;magenta;green;cyan;yellow
```

The order of the named colors does not matter.

FAP2PDF Use the FAP2PDF utility to convert FAP files into PDF files. To produce a PDF file from a FAP file, you must also supply an FXR file. The FXR file determines what fonts are used in the PDF files the utility produces.

The utility looks in the PDF printer control group in your FSISYS.INI file for options that tell it how to create the PDF file. These options let you include bookmarks, embed fonts, use internal fonts, and so on.

NOTE You can use the /INI parameter to specify the INI file that contains the PDF printer control group and you can use the /P parameter to specify the actual name of the PDF printer control group. For information about the INI options you can use to customize PDF output, see Using the PDF Print Driver.

Syntax FAP2PDF /I /X /INI /P /VF

Parameter Description

/I	Enter the names of the FAP files you want to convert into PDF files. You can use wildcard characters (*). You do not have to specify the .FAP extension.
/X	Enter the name of the FXR file you want to use. The FXR file determines which fonts are used in the PDF file. You do not have to specify the .FXR extension.
/INI	(optional) Enter the name of the INI file that contains PDF settings. The default is FSISYS.INI.
/P	(optional) Enter the name of the PDF PrtType control group in the FSISYS.INI file. The default is PDF.
/VF	(optional) Include this parameter if you want to template fields.

If the FXR and INI files specify that you want the fonts embedded, the INI file must contain a FontLib option in the MasterResource control group that specifies the directory of the TrueType or PostScript fonts.

NOTE To see a list of the FAP2PDF parameters, run the utility with no parameters.

FAP2RTF Use the FAP2RTF utility to convert a FAP file into an RTF file that can be used, for instance in a word processor such as Microsoft Word.

Syntax `fap2rtf /i /x /ini /o`

Parameter Description

<code>/i</code>	Enter the name of the FAP file you want to convert.
<code>/x</code>	Enter the name of the corresponding FXR file.
<code>/ini</code>	Enter the name of the INI file which contains the options for this utility.
<code>/o</code>	(Optional) Enter the name for the output RTF file. If you omit this option, the system uses the name of the FAP file and adds the RTF extension.

Here is an example:

```
Fap2rtf /i=qladdr.fap /x=rell03sm.fxr /ini=fapcomp.ini /o=test.rtf
```

This example shows how you can use wild cards:

```
Fap2rtf /i=*.fap /x=rell03sm.fxr /ini=fapcomp.ini
```

Wild cards are not supported if you also use the `/o` parameter.

INI File Settings

The following control groups and options are required for the FAP2RTF utility.

```
< Printer >
PrtType      = RTF
```

Option Description

PrtType	Enter a name such as RTF.
----------------	---------------------------

```
< Printers >
PrtType      = RTF
```

Option Description

PrtType	Enter the name you entered for this option in the Printer control group.
----------------	--

```
< PrtType:RTF >
Module          = RTFW32
PrintFunc       = RTFPrint
Device          = c:\Frame.RTF
PageNumbers     = Yes
AllowInput      = Yes
EmptyHeaders    = Yes
EmptyFooters    = Yes
WriteFrames     = No
BmSub           = Yes
BmSubChar       = -
TemplateFields  = Yes,Enabled
SendColor       = Yes,Enabled
```

Option	Description:
Module	The DLL name of the RTF driver.
PrintFunc	Enter the function name of the print function.
Device	Enter the name of output file when printing from Image Editor.
PageNumbers	Enter Yes to print page numbers on each page. The default is No.
AllowInput	Enter Yes to enable form fields. The default is No.
EmptyHeaders	Enter Yes if you want empty headers in the RTF file. The default is No.
EmptyFooters	Enter Yes if you want empty footers in the RTF file. The default is No.
WriteFrame	Enter No if frames are not required. The default is Yes.
BmSub	Enter No if you <i>do not</i> want to replace invalid characters. The default is Yes.
BmSubChar	Enter character to use when invalid characters are substituted. The default is the underscore (_).
TemplateFields	The default, Yes, tells the system to test print Xs in variable fields. If you also include Enabled (Yes,Enabled), the Template Variable Fields field in Image Editor is checked.
SendColor	Enter Yes to print in color. The default is No. If you also include Enabled (Yes,Enabled), the Send Color field in Image Editor is checked.

If you have chosen to allow form fields, you may also need to include the WordTimeFormats and WordDateFormats control groups. You can use these control groups in case you are using a time or date format in Image Editor that has no equivalent in Word. The following groups and options let you map a Skywire Software format to a Word format.

```
< WordTimeFormats >
    hh:mm XM =
< DateTimeFormats >
    bD/bM/YY =
```

To the left of the equals sign, you list the Skywire Software format used on the image. To the right, you list the Word format you want to use.

FDT2CFA Use the FDT2CFA utility to convert all of the FAP files listed in a FORM.DAT file into compiled FAP files. By pre-compiling your FAP files, you can speed processing.

NOTE The version of the system you use to compile the FAP and FXR files must be the same version you will use when running Documaker Server. Furthermore, the platforms must also match. For instance, if you compile the FAP and FXR files on version 11.0 for Windows, to use them in Documaker Server (GenTrn, GenData, GenPrint), you must run version 11.0 for Windows of Documaker Server.

Program names

Windows FDT2CFAW.EXE

Syntax FDT2CFAW /INI /I /F /L /O

Parameter Description

/INI	The input FSIUSER.INI file.
/I	The input FORM.DAT file.
/F	The input font cross-reference (FXR) file.
/L	The input FAP file library.
/O	The output CFA file library.

This utility lets you compile all of the FAP files in a FORM.DAT file at a time. To compile individual FAP files, see [FAP2CFA on page 67](#). To convert a CFA or CFX file back into a FAP or FXR file, see [CFA2FAP on page 53](#).

Make sure the CompiledFAP option is set to **Yes** (the default) in the RunMode control group of your INI files before you run the system using precompiled FAP files.

In addition, use the File, Library Setup option (or edit your INI files) to specify the path for the CompLib, which is where the system will look for the compiled files. In your INI files, you will find this setting in the following control group:

```
< MasterResource >
  CompLib = (directory the CFA and CFX files are stored in)
```

For MVS systems, you can specify a DD name, such as DD:COMPLIB()

NOTE If you are going to use pre-compiled FAP files, you must also use compiled FXR files.

FDT2DB

Use this utility to create a cross-referenced database of your master resource library (MRL) forms, images, and fields. You can use the resulting database to query this information from within any appropriate database tool. For instance, you can use this information to:

- Build basic reports, such as a field usage report
- Design import/export files for interfacing to other systems
- Create files for use with Transall

The utility uses the FORM.DAT or BDF (Business Definition File) file to extract resource information into a database file, with tables for the library, form group keys, global recipients, form groups, forms, images, linked recipients, and fields.

NOTE Rules are not converted.

The utility reads the entire MRL, but if the resource is not listed in the FORM.DAT or BDF file, the utility does not include it.

Program names

Windows FDT2DB.EXE

Syntax

FDT2DB /? /INI /I /BDF /Purge

Parameter Description

/?	(Optional) Prints to the console all command line options and defaults. No processing occurs when you include this parameter.
/INI	(Optional) Tells the utility which INI file to use. The default is the FSIUSER.INI file. The information in the INI file tells the utility where to find the master resource library (MRL).
/I	(Optional) The name of the input file. If you omit this parameter, the utility defaults to FORM.DAT. If you enter a file name but omit the extension, the utility defaults to DAT DAT — unless you include the /BDF parameter, in which case it defaults to BDF BDF . If you omit the extension, include the period at the end of the file name.
/BDF	(Optional) Tells the utility to use the Business Definition (BDF) file instead of the FORM.DAT file. You must include this parameter if you want the utility to use a BDF file when it converts resources. Note: If you plan to use the output in the PPS Reporting Tool, do not include this parameter.
/Purge	(Optional) Tells the utility to remove the current table data before it repopulates the database with data from the BDF or FORM.DAT file.

You can run this utility multiple times. Be aware, however, that if you run it multiple times, data may be duplicated and you may have unnecessary records in the database. To remove unnecessary records, include the /Purge parameter. This tells the utility to remove all records from the database before opening the input file. After the purge, only records from the input file are in the database file.

Specifying the database Use the following INI options to specify the database you want to use:

```
< DBHandlers >
    DBHandler = ODBC
```

You cannot run this utility without the DBHandler option. The DBHandlers control group defines the ODBC available to the system. With this option, you can name the ODBC anything you want.

```
< DBHandler:ODBC >
    Server = RPEX1
```

The Server option is referenced by the DBHandler option. The utility searches for the DBHandlers:XXXX (XXXX being the selected ODBC) for Server option. If you omit these options and choose to use a blank database, the utility uses the default table DFDs.

Creating a Database

You can create an ODBC database and use Windows' Control Panel to open the ODBC applet and add a new database connection by following these steps:

NOTE To use the new database, remember to set up the Server option in the INI file.

- 1 Open Windows' Control Panel. Double-click the ODBC Data Sources icon.
- 2 On the User DSN page (the first page), click Add. The next page that appears should have the Microsoft Access Driver selected. If not, select it and click Finish.
- 3 On the next window, enter a name for your database in the Data Source Name field. Then click Create.
- 4 On the File window, move to the directory where you want the database created. Enter the database file name. This does not have to be the same name as the Data Source Name. When you finish, click Ok. You should now see the database name above the buttons in the Data Source Name window. Click OK again.

Now you see the list of databases window again and your new database should appear in the list. Click Ok to close the window.

NOTE For Windows 2000, first select Administrative Tools which takes you to another window that contains the ODBC Data Sources icon.

- 5 Edit the FSIUSER.INI for the new database. Add these lines (assuming its name is *RPEX1*)

```
< DBHandler:ODBC >
    Server = RPEX1
```

The Server option defines the Data Source Name you used to name the database, not the file name of the database.

- 6 Run the utility in the directory where your INI files reside. When it finishes, you should be able to open the database using Access to see that it is populated.

Using an Existing Database

If you use an existing database, make sure the tables are in the following formats:

Library table

Key	Type	Description
LibraryName	Text (32)	Short name to identify a library set. This relates a name to a path where the rest of the tables reside.
LibraryDescription	Text (50)	
LibraryPath	Text (255)	MRL library path

FormGroupKeys table

Key	Type	Description
KeyType	Text (3)	For example, 001 = key 1 (such as Company), 002 = key 2 (such as Line of business).
KeyName	Text (30)	Key name used in the FormGroups table.

GlobalRecipients table

Key	Type	Description
RecipName	Text (30)	Short recipient name.
RecipDescription	Text (50)	Longer description.
RecipCode	Text (20)	Used in some print sorting.
Recip_ID	Number	Unique ID. Used to link recipients.

FormGroups table

Key	Type	Description
GroupName1	Text (30)	Such as a company. Must be an entry in the FormGroupKeys table where KeyType = 001.
GroupName2	Text (30)	Such as a line of business. Must be an entry in the FormGroupKeys table where KeyType = 002.
GroupName3	Text (30)	Such as a recipient. Not yet supported. When supported, this value must be a name in the GlobalRecipients table.
Group_ID	Number	Unique ID. Used to group forms within the Forms table

Forms table

Key	Type	Description
Group_ID	Number	Key from the FormGroups table
FormName	Text (30)	
FormEffectiveDate	Date/time	
FormDescription	Text (50)	
FormOptions	Text (20)	
FormRevInfo	Text (20)	Reserved for future use.
Form_ID	Number	Unique ID. Used to group images in the Images table.

Images table

Key	Type	Description
Form_ID	Number	Key from the Forms table.
FormSequence	Number	The order within the form.
ImageName	Text (30)	The image name.
ImageEffectiveDate	Date/time	
ImageDescription	Text (50)	The image description.
ImageOptions	Text (20)	
ImageRevInfo	Text (20)	Reserved for future use.
Image_ID	Number	Unique ID. Used to group fields in the Fields table.

LinkedRecipients table

Key	Type	Description
RecipScope	Number	Contains 0=image, 1=form, 2=group
Referring_ID	Number	Form_ID, or Image_ID, or Group_ID
Recip_ID	Number	The key from the GlobalRecipients table.
RecipCopyCount	Number	

Fields table

Key	Type	Description
FieldScope	Number	0=image, 1=form, 3=form set
Referring_ID	Number	0=global, else, Form_ID or Image_ID
FieldName	Text (32)	
FieldType	Text (20)	
FieldFormat	Text (20)	
FieldLength	Number	

INI options

Also, if you use an existing database and the tables are already created, make sure you have these INI options set up:

```
< ODBC_FileConvert >
  FrmGrpKy = FormGroupKeys
  GlbRec   = GlobalRecipients
  FrmGrps  = FormGroups
  LnkRcps  = LinkedRecipients
```

These are necessary because some types of databases do not like tables with long names.

FDT2EDL Use this utility to generate EDL output from a FORM.DAT file. You can use this utility to read a FORM.DAT file and convert an master resource library (MRL) into a Documerge EDL file.

NOTE An EDL file serves the same function in the Documerge system as an FDT file does in the DAP system.

Program names

Windows FDT2EDL.EXE

Syntax

FDT2EDL /INI /O /N

Parameter Description

/INI	Enter the path and file name of FSIUSER.INI file.
/O	Enter the drive and path of output directory.
/N	Enter the name of output file you want to create.

For each line in the FORM.DAT file, the images are parsed and a group of entries are written out for each image.

```
Output [*<edlname>] required section header-<edlname> supplied by
command line /N parameter
[ChnDir=<edlpath>\] directory location of chain files, with trailing
backslash
```

The following entries are written for each image:

```
[<membername>]
1+Info=ED=YYYYMMDD,MD=YYYYMMDD,DTN=0,Desc=36 spaces
HiRev=n
1+CHNS=
```

Where...

Component	Description
[*<edlname>]	the output EDL name supplied via command line parameter
[ChnDir=<edlpath>]	the output directory supplied via command line parameter.
[<membername>]	the name of the image file (FAP) for the form in the FORM.DAT file
ED	YYYYMMDD is the current date
MD	YYYYMMDD is the current date
DTN	is always zero

Component	Description
Desc	36 spaces
HiRev	is always 1
1+CHNS	entry with no value

The utility returns success or failure. The following errors may occur:

- Output path was not specified
- Output file name was not specified
- Could not open the FSIUSER.INI file
- Could not open the FORM.DAT file
- Could not open the output file

FIXFNT Use this utility to test and fix Xerox fonts so you can use them with optimized Metacode. If you notice that text prints in the wrong location on the page when a Metacode print stream is optimized but prints correctly when not optimized, you may be using a font that needs to be corrected.

See [METOPT on page 150](#) for more information about optimizing Metacode print streams.

NOTE Docuview, which is part of Docusave Workstation, requires Xerox resources to be padded to fill 512 byte blocks. This includes Xerox fonts. Some old Xerox fonts built with prior versions of the system do not meet this criteria. You can use this utility to read and update those Xerox fonts so they can be used by Docuview.

Program names

Windows FIXFNTW.EXE

Syntax

FIXFNT /I /T

Parameter Description

/I	The name of the FNT file. You can use asterisks (*) as wildcards.
/T	Add this parameter if you only want to test the Xerox font.

Use the /T option to test any Xerox font to see if it can be used with Metacode optimization.

Once this utility finishes, download the corrected fonts to the printer to eliminate the problem.

Understanding the System

Before version 10.0, logos converted to Xerox fonts contained font specification table entries that prevent them from being used with Metacode optimization. In most cases, this is not an issue if logos are placed on forms. To cause a problem with Metacode optimization, you must use a text label instead of logo for printing a signature or graphic logo and the Xerox signature font must be inserted into the FXR file.

FIXFORM Use the FIXFORM utility to make changes and correct problems in FAP and DDT files. This utility is often used when converting older FAP files into a newer format. You can also use it, for instance, to check DAL script syntax and perform other tasks such as reassigning font IDs, and modifying text rectangle coordinates which define the space reserved for text.

NOTE You can also use Docucreate to perform many of these same tasks. Use the File, Convert, Multiple FAPs option on the main menu.

Program names

Windows FIXFMW32.EXE

Syntax

FIXFMW32 /I /X /O /SHRINK /SS /SX /G/P

Parameter Description

/I	Enter the name of the FAP file.
/X	Enter the name of the font cross-reference (FXR) file.
/O	Enter the name of the output FAP file if different. Omit the extension.
/SHRINK	Include this parameter to shrink the FAP file to a custom size. This parameter tells the utility to remove all of the white space from the bottom, right-hand side of the image. This has the same affect as using the Auto Size button on the Image Editor's Page Properties window.
/SS	Include this parameter to check field DAL script syntax.
/SX	Include this parameter to check field DAL script syntax and attempt to correct errors.
?G	Include this parameter to load global bitmaps (logos).
/P	Enter the name of the INI file.

You can also include these conversion options:

Option	Description
/A	Convert all colored objects to print in color
/C	Combine text elements. The utility defaults to combining text elements if the space between those elements is less than 99% of the width of the space character. You can also specify the percentage. For instance, /C50 tells the utility to combine the text elements into a single text element if the distance between them is less than 50% of the width of the space character.
/M	Map alternate fonts
/K	Map alternate characters